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**Development of a Low-Cost Multimodal
Amperometric Device with Web-Based
Monitoring Interface for Molecular Sensing**

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This is to certify that the thesis entitled “**Development of a Low-Cost Multimodal Amperometric Device with Web-Based Monitoring Interface for Molecular Sensing**” submitted by **Arkya Jyoti Ghosh** (Roll No. 21014003) to the Indian Institute of Technology (BHU), Varanasi, is a record of original research work carried out by him under my supervision and guidance in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology and Master of Technology (Dual Degree)** in **Biochemical Engineering**.

To the best of my knowledge, the matter embodied in this thesis has not been submitted to any other university or institute for the award of any degree or diploma.

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I declare that:

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ABSTRACT

Electrochemical detection methods, particularly amperometry and chronoamperometry, offer exceptional sensitivity, selectivity, and miniaturisation potential for point-of-care biosensing. However, laboratory-grade potentiostats remain prohibitively expensive—commercial instruments such as the PalmSense EmStat series cost approximately ~Rs5,00,000—placing them out of reach for resource-limited research settings. This thesis presents the design, fabrication, firmware development, and multi-analyte validation of a low-cost, portable, web-connected amperometric device aimed at democratising electrochemical sensing.

The hardware platform, designated SUPRANK_V1, is built around the ESP32-WROOM-32E System-on-Chip, which provides a dual-core 240 MHz processor together with integrated 802.11 b/g/n Wi-Fi connectivity. A 16-bit dual-channel digital-to-analogue converter (AD5667R, Analog Devices) provides precise potential control, while a quad-channel precision operational amplifier (AD8608, Analog Devices) implements a transimpedance amplifier (TIA) for current-to-voltage conversion and a voltage follower for reference electrode buffering. Four selectable gain resistors switched via MAX4643/MAX4644 analogue switches enable current measurement across multiple orders of magnitude. Power management is handled by a MCP73831 lithium-polymer charger and dual MIC5205 3.3 V low-dropout regulators with a star-point grounding topology to minimise analogue-digital interference. The printed circuit board was designed in EasyEDA and fabricated by JLCPCB, yielding a total bill-of-materials cost of approximately ~Rs5,000–7,000.

The firmware, developed using the ESP32 Arduino framework, implements a complete chronoamperometry routine with user-configurable applied potential, acquisition duration, sampling interval, and current range. A single-page web application served directly from the microcontroller enables wireless real-time visualisation via Chart.js, with export options in CSV and JSON formats. An enclosure was designed in Autodesk Fusion 360 and printed in PLA using a Prusa Mini FDM printer.

Electrochemical performance was validated using four analyte-electrode systems: (i) ferri/ferrocyanide $[\text{Fe}(\text{CN})_6]^{3-/4-}$ as an outer-sphere redox probe for basic device characterisation; (ii) ruthenium hexamine $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ as a surface-insensitive probe; (iii) hydrogen peroxide (H_2O_2) at gold screen-printed electrodes; and (iv) p-cresol, a clinically relevant uremic toxin, over a concentration range spanning $1 \times 10^{-15} \text{ M}$ – $1 \times 10^{-2} \text{ M}$. Cottrell analysis of chronoamperometric transients confirmed the device's ability to resolve diffusion-controlled faradaic currents consistent with literature-reported values. The system achieves sub-femtomolar detection of p-cresol, demonstrating ultra-high sensitivity competitive with laboratory instruments at a fraction of the cost.

Keywords: amperometric biosensor, chronoamperometry, ESP32, low-cost potentiostat, web-based monitoring, p-cresol, hydrogen peroxide, point-of-care diagnostics.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADC	Analogue-to-Digital Converter
AGND	Analogue Ground
API	Application Programming Interface
BOM	Bill of Materials
CA	Chronoamperometry
CE	Counter Electrode
CV	Cyclic Voltammetry
DAC	Digital-to-Analogue Converter
DET	Direct Electron Transfer
DGND	Digital Ground
DPV	Differential Pulse Voltammetry
EIS	Electrochemical Impedance Spectroscopy
ESP32	Espressif Systems MCU with integrated Wi-Fi and Blue-tooth
FDM	Fused Deposition Modelling
FTDI	Future Technology Devices International (USB-to-UART bridge)
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
I ² C	Inter-Integrated Circuit (serial communication protocol)
IoT	Internet of Things
JSON	JavaScript Object Notation
LDO	Low-Dropout Regulator
LiPo	Lithium Polymer
LOD	Limit of Detection
LOQ	Limit of Quantification
MCU	Microcontroller Unit
MSOP	Mini Small Outline Package
OCP	Open-Circuit Potential
PBS	Phosphate-Buffered Saline
PCB	Printed Circuit Board
PLA	Polylactic Acid

Abbreviation	Full Form
POC	Point-of-Care
RE	Reference Electrode
RSD	Relative Standard Deviation
SoC	System-on-Chip
SPE	Screen-Printed Electrode
SPI	Serial Peripheral Interface
SPIFFS	Serial Peripheral Interface Flash File System
TIA	Transimpedance Amplifier
TOF	Time of Flight
UART	Universal Asynchronous Receiver-Transmitter
USB	Universal Serial Bus
WE	Working Electrode
Wi-Fi	Wireless Fidelity (IEEE 802.11)
WS	WebSocket

Chapter 1

INTRODUCTION

1.1 Background and Motivation

The global burden of non-communicable diseases, environmental contaminants, and food safety hazards has created an urgent demand for rapid, sensitive, and affordable analytical tools that can function outside the confines of centralised laboratories [1]. Electrochemical biosensors—devices that transduce biochemical recognition events into measurable electrical signals—have emerged as one of the most promising platforms to address this challenge. Their inherent miniaturisability, low power consumption, compatibility with batch microfabrication, and potential for real-time digital readout make them ideally suited for point-of-care (POC) diagnostics, environmental monitoring, and food quality assessment [2, 3].

Among the various electroanalytical techniques, *amperometry* and *chronoamperometry* (CA) occupy a privileged position. Amperometric sensors operate by holding the working electrode at a fixed potential and measuring the resulting faradaic current, which is proportional to the analyte concentration. Chronoamperometry extends this principle to the time domain, providing rich mechanistic information about electrode kinetics, diffusion coefficients, and adsorption phenomena through the analysis of transient current–time (*i*–*t*) profiles [4]. These techniques are employed in a wide range of important applications, from clinical glucose monitoring to heavy metal detection and enzyme-linked immunoassays [5, 6].

Despite their potential, the widespread deployment of amperometric biosensors has been hindered by the high cost of laboratory-grade potentiostats—the electronic instruments that precisely control and measure electrochemical cell potentials and currents. Commercial systems from manufacturers such as PalmSense (Netherlands), Gamry Instruments (USA), Bio-Logic (France), and CH Instruments (USA) are highly capable but cost between \$1,000 and \$30,000 per unit, which translates to ~Rs80,000–Rs25,00,000 in the Indian context. Even entry-level portable units such as the PalmSense EmStat Pico are priced at approximately ~Rs5,00,000, placing them beyond the reach of resource-limited research groups, undergraduate teaching laboratories, field monitoring applications, and small-scale clinical facilities in low- and middle-income countries [7, 8].

This cost barrier has motivated a growing community of researchers, engineers, and makers to develop open-source and low-cost potentiostat platforms [9, 10]. Early efforts such as the CheapStat, developed by Rowe *et al.* at the University of California, Santa Bar-

bara, demonstrated that a fully functional three-electrode potentiostat could be assembled for under \$80 using off-the-shelf components and open-source firmware [8]. Subsequent systems—including DStat [10], UWED [7], RODEOSTAT, and Sensit Smart—have progressively improved performance and ease of use. However, most of these systems rely on tethered USB connections to a host computer for data acquisition and visualisation, limiting their portability and ease of use in field settings.

The advent of powerful, low-cost system-on-chip (SoC) microcontrollers with integrated wireless connectivity—most notably the Espressif ESP32 family—has opened entirely new possibilities for truly portable, self-contained electrochemical instruments [11, 12]. The ESP32-WROOM-32E module integrates a dual-core 240 MHz Xtensa LX6 processor, 4 MB flash memory, 520 kB SRAM, and a full IEEE 802.11 b/g/n Wi-Fi stack, all in a compact 18×20 mm package costing under \$5. When combined with a high-resolution digital-to-analogue converter (DAC), precision operational amplifiers, and range-switching analogue circuitry, the ESP32 forms the nucleus of a potentiostat that can serve its own web-based monitoring interface directly to any Wi-Fi-enabled device—eliminating the need for dedicated software installation or physical tethering.

1.2 Significance of Target Analytes

The validation of any new analytical platform requires a careful selection of analyte–electrode pairs that span a range of chemical complexity and biological relevance.

1.2.1 Ferri/Ferrocyanide Redox Couple

The hexacyanoferrate redox couple $[\text{Fe}(\text{CN})_6]^{3-/4-}$ has been used for decades as a benchmark outer-sphere one-electron mediator for electrochemical device characterisation [4]. Its well-established electrochemistry, known diffusion coefficient ($D \approx 7.2 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ in 0.1 M KCl), and insensitivity to most surface conditions make it ideal for validating the potential accuracy and current linearity of a new instrument [13].

1.2.2 Ruthenium Hexamine

Ruthenium hexamine ($[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$) is an outer-sphere one-electron redox probe whose rate constant is largely independent of electrode surface chemistry, making it a complementary standard to ferri/ferrocyanide for diagnosing the electrical (as opposed to chemical) properties of the electrode interface [14]. Its standard potential ($E^0 \approx -0.152 \text{ V}$ vs. Ag/AgCl) is distinct from that of ferri/ferrocyanide, verifying the DAC's ability to address different potential windows.

1.2.3 Hydrogen Peroxide at Gold Electrodes

Hydrogen peroxide (H_2O_2) is arguably the most important transduction species in enzyme-linked amperometric biosensors. Oxidases such as glucose oxidase (GOx), cholesterol oxidase, and lactate oxidase all generate H_2O_2 as a by-product of substrate oxidation, and its amperometric detection provides the analytical signal in a vast number of commercially successful biosensors [15]. Gold (Au) screen-printed electrodes are widely used for H_2O_2 detection owing to their facile electro-oxidation catalysis and surface functionalisation chemistry [16].

1.2.4 p-Cresol

Para-cresol (4-methylphenol, p-cresol) is a phenolic compound produced by the colonic fermentation of aromatic amino acids (tyrosine and phenylalanine) by gut microbiota [17, 18]. In healthy individuals, p-cresol is conjugated in the gut epithelium and liver, predominantly to p-cresyl sulfate (pCS), which is efficiently cleared by the kidneys. In patients with chronic kidney disease (CKD) and end-stage renal disease (ESRD), impaired clearance leads to significant accumulation of pCS and free p-cresol in plasma, where they exert direct cardiotoxic, nephrotoxic, and endothelial effects [19]. Serum p-cresol has consequently been identified as an independent predictor of cardiovascular mortality in dialysis patients, making it a clinically valuable biomarker [20]. The need for sensitive, rapid, and field-deployable detection of p-cresol over a wide dynamic range (femtomolar to millimolar) motivates its selection as a key analyte in the present work.

1.3 Objectives of the Present Work

The overarching aim of this thesis is to develop and rigorously validate a low-cost, portable, web-connected amperometric device that can perform chronoamperometric measurements across multiple analyte–electrode systems. The specific objectives are:

- i. To design a custom printed circuit board (PCB), designated **SUPRANK_V1**, incorporating an ESP32-WROOM-32E microcontroller, a 16-bit dual-channel DAC (AD5667R), a quad-channel precision transimpedance amplifier (AD8608), and selectable gain resistors with analogue switching (MAX4643/MAX4644), implementing star-point grounding and separate analogue/digital power domains.
- ii. To develop firmware on the ESP32 Arduino platform implementing a complete chronoamperometry engine with configurable applied potential, acquisition duration, sampling interval, and current range, and to flash this firmware using an FTDI USB-to-UART programmer.

- iii. To design and deploy a browser-based monitoring interface served directly from the microcontroller over Wi-Fi, providing real-time graphical visualisation, parameter control, and data export in CSV and JSON formats.
- iv. To design a 3D-printed enclosure using Autodesk Fusion 360 and fabricate it using a Prusa Mini FDM printer in PLA, providing a ruggedised and compact housing for the electronics.
- v. To validate the assembled device against established electrochemical standards ($[\text{Fe}(\text{CN})_6]^{3-/4-}$ and $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ in phosphate-buffered saline) and to demonstrate multi-analyte detection capability using H_2O_2 /gold and p-cresol systems.
- vi. To characterise the analytical performance metrics (sensitivity, limit of detection, limit of quantification, linear dynamic range, and repeatability) and to benchmark the device against commercial instruments, demonstrating a cost reduction of two orders of magnitude.

1.4 Scope of the Thesis

The present work is focused exclusively on chronoamperometric (CA) measurements using a three-electrode electrochemical cell comprising a working electrode (WE), a reference electrode (RE), and a counter electrode (CE) in an aqueous electrolyte. While the hardware platform is designed to support additional electroanalytical techniques (e.g. cyclic voltammetry, differential pulse voltammetry) through firmware updates, this thesis addresses only CA. The device operates in the potential range achievable by the AD5667R DAC with the op-amp gain network implemented on the PCB. All experiments are conducted in a laboratory benchtop setting; the clinical or in-field deployment of the device is identified as future work.

1.5 Organisation of the Thesis

The remainder of this thesis is organised as follows:

Chapter 2 presents a comprehensive review of the literature relevant to the present work, covering the fundamentals of electrochemical sensing and chronoamperometry, commercial and open-source potentiostat platforms, ESP32-based IoT sensing, and the biochemistry and analytical detection of the target analytes.

Chapter 3 describes the materials and methods employed, including the detailed hardware design (circuit schematic, PCB layout, component selection rationale), firmware development, web interface implementation, 3D enclosure fabrication, and experimental protocols for chemical preparation and electrochemical measurements.

Chapter 4 presents and discusses the experimental results: PCB validation, device characterisation, chronoamperometric responses to each analyte, Cottrell analysis, calibration curves, and a comprehensive cost analysis.

Chapter 5 draws conclusions from the work, identifies its limitations, and proposes directions for future research including expanded analyte panels, additional electroanalytical techniques, and clinical validation studies.

Chapter 2

LITERATURE REVIEW

2.1 Fundamentals of Electrochemical Sensing

2.1.1 The Three-Electrode Electrochemical Cell

An electrochemical cell is a system in which chemical energy is interconverted with electrical energy at the interface between an electronic conductor (electrode) and an ionic conductor (electrolyte). For analytical applications involving controlled-potential experiments, the *three-electrode cell* configuration is universally employed to decouple the potential control function from the current measurement function [4].

The three electrodes are:

- **Working electrode (WE):** The electrode at whose surface the analytical reaction of interest takes place. The potential of the WE is controlled with respect to the RE. Common WE materials include glassy carbon, gold, platinum, indium tin oxide (ITO), and carbon paste or screen-printed carbon.
- **Reference electrode (RE):** An electrode of stable, well-defined potential against which the WE potential is measured. Common reference electrodes include the silver/silver chloride (Ag/AgCl , $E = +0.197 \text{ V}$ vs. SHE in 3 M KCl), saturated calomel electrode (SCE, $+0.241 \text{ V}$ vs. SHE), and the standard hydrogen electrode (SHE). The RE passes negligible current.
- **Counter electrode (CE):** Also called the auxiliary electrode, the CE completes the electrical circuit and carries the faradaic current. It is typically a platinum wire or a large-area carbon rod. By using a three-electrode configuration with a potentiostat, the uncompensated solution resistance (R_u) between the RE and WE is minimised, enabling accurate potential control.

The potentiostat is the electronic instrument that controls $E_{\text{WE}} - E_{\text{RE}}$ and simultaneously measures the resulting current [21]. The classical potentiostat circuit employs an electrometer amplifier (high-impedance voltage follower) to sense the RE potential, a summing (control) amplifier to force the WE potential to the desired set-point by driving current through the CE, and a current-to-voltage converter (transimpedance amplifier, TIA) to measure the cell current.

2.1.2 Faradaic and Non-Faradaic Currents

The total current flowing in an electrochemical cell has two components:

$$i_{\text{total}}(t) = i_{\text{F}}(t) + i_{\text{C}}(t) \quad (2.1)$$

where i_{F} is the *faradaic current* arising from electron-transfer reactions at the electrode surface (governed by Faraday's law: $Q = nFN$, where N is the number of moles of electroactive species converted), and i_{C} is the *non-faradaic* (capacitive or charging) current associated with the charging and discharging of the electrical double layer (EDL) at the electrode–solution interface [4]. For analytical purposes the faradaic current carries the analytical signal, while the capacitive current constitutes noise, particularly at short time-scales following a potential step.

2.2 Chronoamperometry: Theory and Analysis

Chronoamperometry involves the application of a square potential step to the working electrode and recording of the resulting current transient as a function of time. Consider a potential step from an initial potential E_i (where no faradaic reaction occurs) to a final potential E_f (where the surface concentration of the electroactive species is driven to zero, i.e. diffusion-limited conditions). For a planar electrode in an unstirred solution, the current–time response is governed by the **Cottrell equation** [22]:

$$i(t) = \frac{nFACD^{1/2}}{\pi^{1/2}t^{1/2}} \quad (2.2)$$

where:

- n = number of electrons transferred per molecule
- F = Faraday's constant (96485 C mol⁻¹)
- A = electrode area (cm²)
- C = bulk concentration of the electroactive species (mol cm⁻³)
- D = diffusion coefficient of the electroactive species (cm² s⁻¹)
- t = time elapsed after the potential step (s)

Equation 2.2 predicts that $i \propto t^{-1/2}$; accordingly, a plot of i versus $t^{-1/2}$ (a *Cottrell plot*) should be linear with slope $m = nFACD^{1/2}/\pi^{1/2}$. This linearity can be exploited in two ways: (i) to extract the diffusion coefficient D of a known species (for device characterisation), and (ii) to construct a calibration curve by plotting m versus C for analyte quantification.

At short times after the potential step, the total current is dominated by the double-layer charging transient:

$$i_C(t) = \frac{\Delta E}{R_u} \exp\left(-\frac{t}{R_u C_{dl}}\right) \quad (2.3)$$

where R_u is the uncompensated resistance and C_{dl} is the double-layer capacitance. After this transient has decayed (typically within a few milliseconds for low-resistance aqueous electrolytes), the Cottrell behaviour dominates [4]. In practice, CA data at early times are discarded and Cottrell analysis is applied to the $t > \tau_{RC}$ portion of the transient.

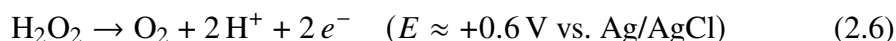
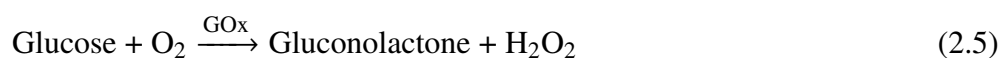
The analytical utility of CA is enhanced by its simplicity (a single potential step), its compatibility with any redox-active analyte, and its straightforward signal-to-concentration relationship. Moreover, the linear i vs. C relationship at a fixed time point t^* provides a simple calibration:

$$i(t^*) = \frac{nFAD^{1/2}}{\pi^{1/2}(t^*)^{1/2}} \cdot C \quad (2.4)$$

enabling quantitative determination of C from a single measurement.

2.3 Amperometric Biosensors

Amperometric biosensors combine the selectivity of a biological recognition element (enzyme, antibody, aptamer, molecularly imprinted polymer, or whole cell) with the sensitivity of electrochemical transduction. The concept was pioneered by Leland Clark Jr. and Champ Lyons in 1962 with their demonstration of the first enzyme electrode: glucose oxidase immobilised on a platinum electrode for glucose detection [23]. This device, the direct antecedent of today's blood glucose monitors, exploited the amperometric detection of H_2O_2 produced enzymatically:



Since this seminal work, amperometric biosensors have been developed for a vast array of targets including cholesterol [24], lactate [25], cancer biomarkers [26], heavy metals, pesticides, and pathogens [1]. The analytical performance metrics used to characterise biosensors include:

- **Sensitivity (S):** the slope of the calibration curve, expressed in $A M^{-1}$ or $\mu A mM^{-1} cm^{-2}$.
- **Limit of detection (LOD):** defined as $3\sigma/S$, where σ is the standard deviation of the blank signal.
- **Limit of quantification (LOQ):** defined as $10\sigma/S$.

- **Linear dynamic range (LDR):** the concentration range over which the calibration is linear.
- **Selectivity:** the ability to distinguish the target analyte from interferents.
- **Stability:** the retention of analytical performance over time.

2.4 Commercial Potentiostats and Their Limitations

High-performance potentiostats from established manufacturers offer exceptional specifications: sub-picoampere current resolution, potential accuracy in the microvolt range, multi-channel capability, and sophisticated software for electrochemical analysis. Table 2.1 summarises representative instruments and their approximate costs.

Table 2.1: Comparison of selected commercial potentiostats and the device developed in this work.

Instrument	Manufacturer	Interface	Portability	Approx. Cost
EmStat Pico	PalmSense, NL	USB/Bluetooth	Portable	~Rs5,00,000
VSP-300	Bio-Logic, FR	USB/Ethernet	Benchtop	~Rs20,00,000
Reference 600	Gamry, USA	USB	Benchtop	~Rs18,00,000
CHI 660E	CH Instruments	USB	Benchtop	~Rs12,00,000
This work	IIT (BHU)	Wi-Fi (Web)	Handheld	~Rs5,000–7,000

The cost of even the most affordable commercial instrument places it out of reach for undergraduate teaching laboratories, field monitoring programmes, and point-of-care settings in resource-limited environments. Furthermore, most commercial potentiostats require dedicated proprietary software running on a computer, precluding their use without a licensed workstation [27].

2.5 Open-Source and Low-Cost Potentiostat Platforms

The availability of inexpensive microcontrollers and precision analogue ICs has enabled a proliferation of open-source potentiostat designs. This section reviews the most influential contributions.

2.5.1 CheapStat

CheapStat, described by Rowe *et al.* in 2011, was one of the first open-hardware potentiostats to gain wide recognition [8]. Built around an Atmel AVR microcontroller (ATmega32U4), it used a resistor ladder DAC for potential control and was capable of cyclic voltammetry,

square wave voltammetry, and anodic stripping. The complete assembly cost was under \$80. However, its potential range was limited (± 1.65 V), its current resolution was modest, and communication was via USB HID, requiring a host PC.

2.5.2 DStat

The DStat potentiostat, introduced by Dryden and Wheeler [10], represented a significant improvement. Using a PSoC 5LP microcontroller and a precision DAC, DStat achieved ± 1.5 V potential range, 1 nA current resolution, and a software-configurable gain transimpedance amplifier. The open-source Python client (dstatcom) provided a polished GUI. DStat requires USB connection to a computer for operation, limiting its portability.

2.5.3 RODEOSTAT

RODEOSTAT, developed by IO Rodeo, is an open-source potentiostat based on the Teensy 3.2 microcontroller. It supports CV, CA, and linear sweep voltammetry, with a web-based control interface via USB serial. Performance is competitive with entry-level commercial instruments, with a cost around \$150.

2.5.4 UWED

Ainla *et al.* [7] presented the Unattended Wireless Electrochemical Detector (UWED), an early attempt at a truly wireless potentiostat using Bluetooth Low Energy (BLE). The UWED demonstrated that wireless communication was feasible for electrochemical data acquisition, though it was limited to BLE range and required a smartphone app.

2.5.5 ESP32-Based Systems

More recently, the ESP32's combination of processing power and Wi-Fi has attracted attention for electrochemical applications. Jenkins *et al.* [12] demonstrated a field-deployable sensor node using ESP32 for heavy metal detection. Wan *et al.* [28] reported an ESP32-based glucose sensor with a web server interface. However, these implementations either lacked precision DAC-based potential control or did not provide a multi-range TIA, limiting their current resolution and dynamic range. The device described in this thesis addresses these gaps by integrating a 16-bit DAC (AD5667R) and a quad-channel precision TIA (AD8608) with selectable gain, all implemented on a custom PCB.

2.6 IoT and Web-Based Sensing Platforms

The Internet of Things (IoT) paradigm—in which physical sensors communicate via IP networks—has transformative potential for electrochemical sensing [29]. Key enabling

technologies include:

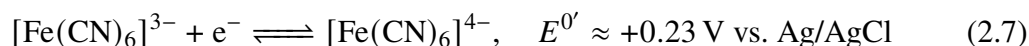
- **ESP32 AsyncWebServer:** An asynchronous HTTP server library that enables the ESP32 to serve static web pages and handle multiple concurrent HTTP requests without blocking the firmware.
- **WebSockets (RFC 6455):** A full-duplex communication protocol over a single TCP connection. WebSocket enables the microcontroller to push real-time data to the browser at each sampling interval, enabling live graph updates without page refresh.
- **SPIFFS / LittleFS:** Flash file systems on the ESP32 that allow static web assets (HTML, CSS, JavaScript) to be stored in flash memory and served on demand.
- **Chart.js:** An open-source JavaScript library for responsive, interactive chart rendering in the browser, requiring no server-side computation.

This architecture allows a smartphone, tablet, or laptop to function as the display and control interface for the potentiostat without any dedicated software installation, making the system truly plug-and-play for end users [30].

2.7 Detection of Target Analytes

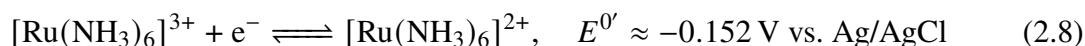
2.7.1 Ferri/Ferrocyanide and Ruthenium Hexamine as Electrochemical Standards

The hexacyanoferrate couple $[\text{Fe}(\text{CN})_6]^{3-/4-}$ undergoes a simple, reversible, outer-sphere one-electron transfer:



Its diffusion coefficient in 0.1 M KCl at 25 °C is $D_{\text{ox}} = 7.2 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ and $D_{\text{red}} = 6.7 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ [13]. Its high solubility, chemical stability, and well-characterised kinetics make it the de facto standard for potentiostat validation.

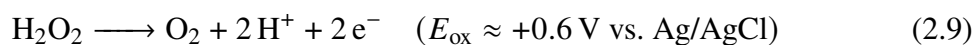
Ruthenium hexamine ($[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$) undergoes a reversible outer-sphere single-electron reduction:



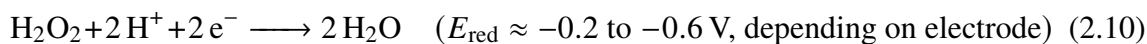
Its electrode kinetics are among the fastest known ($k^0 > 1 \text{ cm s}^{-1}$ on many electrodes) and are essentially independent of surface chemistry [14], providing a complementary validation of the device's potential control at negative potentials.

2.7.2 Hydrogen Peroxide Detection at Gold Electrodes

Hydrogen peroxide can be detected electrochemically by either oxidation:



or reduction:

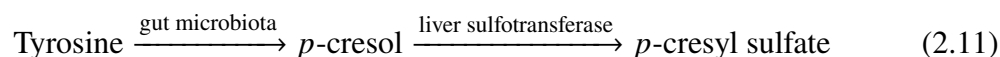


Gold electrodes are particularly attractive for H_2O_2 oxidation because they exhibit lower overpotentials than carbon and are less susceptible to surface fouling than platinum [16]. Screen-printed gold electrodes (SPGEs) offer additional advantages of disposability and low cost. The electrocatalytic oxidation of H_2O_2 at gold surfaces has been exploited in numerous enzyme-based biosensors, and its detection with the present device validates the platform for downstream horseradish peroxidase (HRP)-based sandwich immunoassay architectures [5].

2.7.3 p-Cresol: Biochemistry, Clinical Significance, and Detection

Biochemistry and Metabolic Fate

p-Cresol (4-methylphenol, $\text{MW} = 108.14 \text{ g mol}^{-1}$) is a phenolic compound produced exclusively by anaerobic gut bacteria through the deamination and decarboxylation of the aromatic amino acid tyrosine [17]:



In individuals with normal kidney function, p-cresyl sulfate (pCS) is efficiently cleared by the kidneys and excreted in urine. In CKD patients, particularly those on haemodialysis, pCS accumulates dramatically in plasma, reaching concentrations up to $200 \mu\text{M}$ —several-fold above normal ($<10 \mu\text{M}$) [17, 18].

Clinical Significance

Multiple prospective cohort studies have established pCS and free p-cresol as independent risk factors for cardiovascular disease and all-cause mortality in dialysis patients [19, 20]. p-Cresol also disrupts endothelial cell function, inhibits leukocyte degranulation, and impairs phagocytic killing of bacteria [17]. Its quantification in clinical samples is therefore valuable for CKD staging, dialysis adequacy assessment, and patient risk stratification.

Electrochemical Detection of p-Cresol

p-Cresol is electrochemically oxidisable via a $2e^-/2H^+$ mechanism at moderate anodic potentials [5]. Its phenolic hydroxyl group undergoes irreversible oxidation at carbon or gold electrodes at $\approx +0.6$ – $+0.8$ V vs. Ag/AgCl, generating a quinone product. This direct electrochemistry can be used for label-free amperometric detection without enzymatic amplification. However, the high detection potential overlaps with electrochemical oxidation of common interferents such as uric acid and ascorbic acid, necessitating either selective electrode modification or the use of differential pulse voltammetry. Alternatively, enzyme-based (phenol oxidase, tyrosinase, laccase) architectures exploit the enzymatic oxidation of p-cresol to generate a catechol or quinone product for amperometric transduction [16].

Recent years have seen a surge in reports on ultra-sensitive p-cresol biosensors employing nanomaterial-modified electrodes [26]. Nanostructured gold, reduced graphene oxide, carbon quantum dots, and molecularly imprinted polymers (MIPs) have all been employed to extend the LDR to femtomolar concentrations and achieve LODs in the sub-picomolar range, exploiting signal amplification through increased electroactive surface area and selective analyte pre-concentration. The present work aims to exploit such an electrode platform in conjunction with the custom amperometric device to achieve comparable sensitivity.

2.8 Circuit Design Considerations for Low-Noise Potentiostats

2.8.1 Transimpedance Amplifier Design

The TIA converts the cell current i_{cell} to a voltage $V_{\text{out}} = -i_{\text{cell}} \cdot R_f$, where R_f is the feedback resistor [31]. Key design considerations include:

- **Input bias current:** The op-amp's input bias current adds an offset to the measured cell current. Precision op-amps such as the AD8608 ($\max I_{\text{bias}} = 1$ nA) are essential for low-current measurements.
- **Gain-bandwidth product:** High R_f values increase sensitivity but reduce the TIA bandwidth. A bandwidth of several hundred Hz is sufficient for CA experiments.
- **Stability:** Feedback capacitor C_f (typically 10–100 pF) in parallel with R_f is required to prevent oscillation when measuring high-impedance cells.
- **Noise:** Johnson noise from R_f sets the noise floor: $v_n = \sqrt{4kTR_f\Delta f}$, where k is Boltzmann's constant and Δf is the measurement bandwidth.

2.8.2 Ground Plane Management

Analogue and digital circuits sharing a ground plane can suffer from digital switching noise (from the microcontroller and Wi-Fi transceiver) coupling into the sensitive analogue signal chain. The star-point (single-point) grounding topology separates AGND from DGND, connecting them only at a single low-impedance node near the power supply decoupling capacitors [31]. This prevents digital return currents from flowing through the analogue ground plane, reducing interference to the sub-microampere level.

2.8.3 Power Supply Decoupling

Each analogue IC is decoupled with a 100 nF ceramic capacitor placed as close as possible to the supply pins, supplemented by a 10 μ F bulk capacitor on the supply rail. This suppresses high-frequency supply noise and protects sensitive analogue ICs from transient disturbances caused by the ESP32's Wi-Fi radio transmissions, which can draw instantaneous currents of up to 240 mA.

2.9 Summary

This review has established that:

1. Chronoamperometry at a three-electrode cell provides quantitative analyte concentration information via the Cottrell equation.
2. Commercial potentiostats are prohibitively expensive for widespread POC or teaching use.
3. Existing open-source instruments require tethered PC connections, limiting field utility.
4. The ESP32 SoC with integrated Wi-Fi, combined with a precision DAC and TIA, is an enabling technology for self-contained web-connected potentiostats.
5. p-Cresol is a clinically important uremic toxin for which an ultra-sensitive portable detection platform is urgently needed.
6. Careful PCB layout (star-ground, decoupling) is essential for low-noise electrochemical measurements.

These findings directly motivate the design and implementation described in Chapter 3.

Chapter 3

MATERIALS AND METHODS

3.1 Overview of the System Architecture

The amperometric device, designated **SUPRANK_V1**, is a self-contained, portable instrument that integrates hardware, firmware, and software into a single handheld unit. Figure 3.1 shows a 3D render of the fabricated SUPRANK_V1 PCB. The system comprises four major subsystems: (i) the electrochemical front-end (potential control and current measurement circuits); (ii) the microcontroller unit (ESP32-WROOM-32E) responsible for orchestrating the experiment, digitising the TIA output, and hosting the web server; (iii) the power management subsystem (battery charger, LDO regulators); and (iv) the web-based user interface accessible from any Wi-Fi-enabled device.

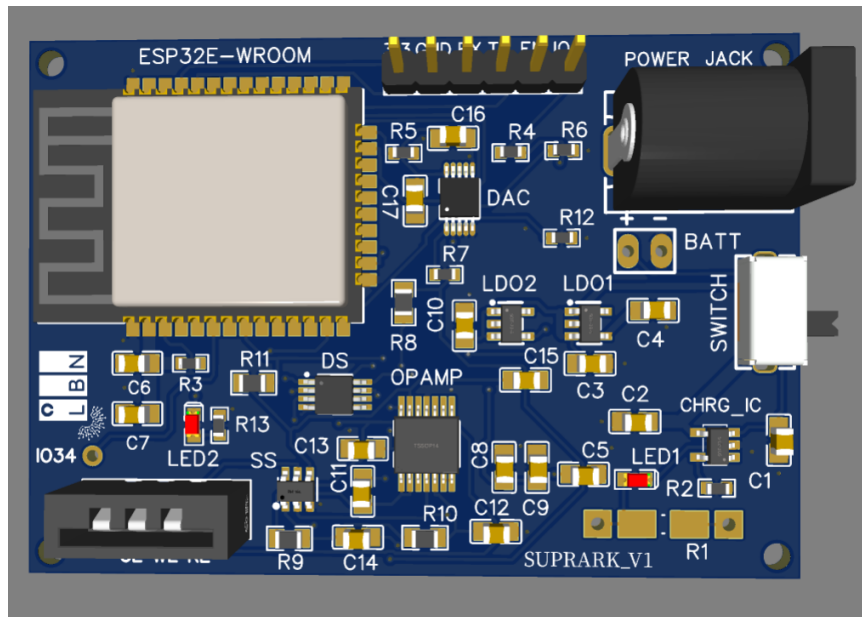


Figure 3.1: 3D render of the SUPRANK_V1 PCB showing the ESP32-WROOM-32E module (top-left), AD5667R DAC, AD8608 quad op-amp, dual LDO regulators, MCP73831 charge IC, power jack, battery connector, slide switch, and status LEDs.

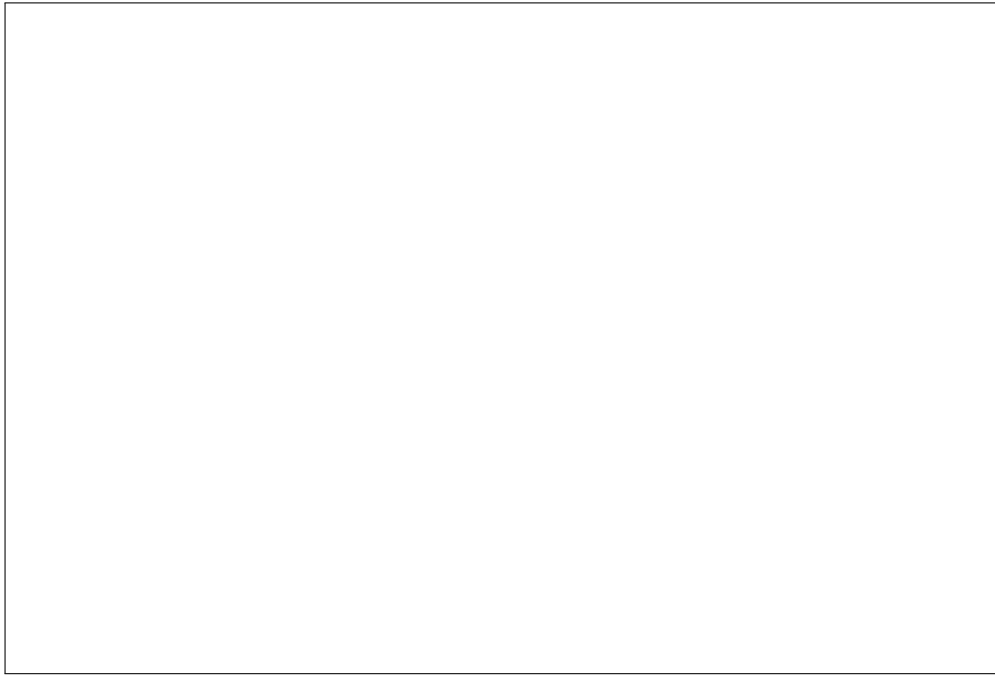


Figure 3.2: Block diagram of the SUPRANK_V1 system showing the four major subsystems: electrochemical front-end, ESP32 MCU, power management, and web-based user interface. Arrows indicate data flow and control signals.

3.2 Hardware Design

3.2.1 Microcontroller: ESP32-WROOM-32E

The central processing unit of the device is the **ESP32-WROOM-32E** module (Espressif Systems, LCSC part C701341), which houses the ESP32-D0WD-V3 dual-core 32-bit Xtensa LX6 CPU operating at up to 240 MHz. The module provides 4 MB of embedded SPI flash memory, 520 kB of SRAM, and integrates a complete IEEE 802.11 b/g/n (2.4 GHz) Wi-Fi radio with an on-board PCB trace antenna. It supports 34 programmable GPIO pins, two 12-bit ADC arrays (SAR ADC, up to 18 channels), multiple UART, SPI, I²C, and I²S peripherals, and a hardware PWM controller. The module operates from a 3.3 V supply and is rated at ~80 mA average and up to 240 mA peak during Wi-Fi transmission.

In the present design, the ESP32 performs three critical functions:

1. **DAC control:** Issues SPI commands to the AD5667R DAC to set the cell potential.
2. **ADC measurement:** Samples the TIA output voltage through an internal 12-bit SAR ADC channel.
3. **Web server:** Hosts the AsyncWebServer and WebSocket server, streaming real-time data to the connected client.

3.2.2 Potential Control: AD5667R 16-bit DAC

Precise and stable potential control is the most critical requirement of a potentiostat. The **AD5667R** (Analog Devices, LCSC part C650006) is a dual-channel, 16-bit, voltage-output DAC in an MSOP-10 package. Its key specifications are:

- Resolution: 16 bits (65536 codes over a 2.5 V or 5 V reference span)
- INL/DNL: ± 1 LSB (max)
- Output voltage range: 0 to V_{ref} (unipolar) or $\pm V_{\text{ref}}/2$ (with op-amp inversion)
- Communication: I²C or SPI (hardware-selectable via pin)
- Power supply: 2.7–5.5 V
- On-chip 2.5 V reference: yes (enabled by default in firmware)

Channel A of the DAC generates the desired electrode potential E_{set} . The raw DAC output (0–2.5 V) is level-shifted and amplified by the op-amp gain network (resistors R8–R11; see Section 3.2.3) to achieve the required potential range. Channel B of the DAC is reserved for future use (e.g. bipotentiostat operation).

The relationship between the digital code D and the output potential E_{set} after the gain network is:

$$E_{\text{set}} = \left(\frac{D}{2^{16} - 1} \right) \cdot V_{\text{FS}} \cdot G_{\text{amp}} \quad (3.1)$$

where V_{FS} is the DAC full-scale voltage (2.5 V) and G_{amp} is the op-amp gain factor. With the component values chosen (see Section 3.2.3), this achieves a potential resolution of ~ 0.1 mV per DAC step over the full operating range.

3.2.3 Current Measurement: AD8608 Quad Precision Op-Amp and TIA

The **AD8608ARUZ** (Analog Devices, LCSC part C578776) is a quad-channel, single-supply precision CMOS operational amplifier in a TSSOP-14 package. Its salient specifications relevant to TIA design are:

- Input offset voltage: $V_{\text{os}} = 65 \mu\text{V}$ (typical), $250 \mu\text{V}$ (max)
- Input bias current: $I_{\text{b}} = 1 \text{ pA}$ (typical)
- Gain-bandwidth product (GBW): 10 MHz
- Rail-to-rail input and output
- Supply voltage: 2.7–5 V (single supply)

- Noise: $e_n = 8 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz

The ultra-low input bias current (1 pA typical) is critical: bias current directly adds to the measured cell current, introducing a fixed offset. At nano-ampere current levels, an op-amp with $I_b = 1 \text{ nA}$ (typical JFET op-amp) would introduce an intolerable $\sim 100\%$ offset.

The four op-amp stages within the AD8608 are allocated as follows:

- **U11-A:** Voltage follower for the reference electrode (RE buffer). The high input impedance ensures negligible current draw from the RE, maintaining a stable reference potential.
- **U11-B:** Control amplifier (error amplifier). Compares the desired potential (from the DAC, buffered through U11-C) with the measured RE potential and drives the counter electrode (CE) to minimise the error.
- **U11-C:** Unity-gain buffer for the DAC output.
- **U11-D:** Transimpedance amplifier. The working electrode is connected to the inverting input (virtual ground). The gain-switching feedback resistor (R_f , selected by the MAX4643/MAX4644 switches; see Section 3.2.4) converts the cell current to a proportional output voltage: $V_{\text{TIA}} = -i_{\text{cell}} \cdot R_f$.

Gain Resistor Values and Current Ranges

The TIA feedback resistor R_f is selected from four values by the analogue switch network, providing the following current ranges:

Table 3.1: TIA gain resistors and corresponding current ranges.

Range	R_f (k Ω)	Full-scale current	Resolution (12-bit ADC)
1	845 (R9)	0 μA –1 μA	0.24 nA
2	8.06 (R11)	0 μA –10 μA	2.4 nA
3	16.2 (R8)	0 μA –50 μA	12 nA
4	29.4 (R10)	0 μA –100 μA	24 nA

The full-scale current is defined by $I_{\text{max}} = V_{\text{DD}}/R_f$ (3.3 V supply). Range 1 is used for low-concentration analyte measurements, while Range 4 is used for high-concentration validation with ferri/ferrocyanide.

3.2.4 Range Switching: MAX4643 and MAX4644 Analogue Switches

The four TIA feedback resistors are selected by a combination of one **MAX4643EUA+** dual-single-pole double-throw (SPDT) analogue switch (Analog Devices/Maxim, LCSC

part C1546042) and one **TPMAX4644EUT+T** single-SPDT switch (Tech Public, LCSC part C4244903). These switches have:

- On-resistance: $R_{\text{on}} \approx 4.5 \Omega$ (negligible compared to R_f in $\text{k}\Omega$ range)
- Off-isolation: $>100 \text{ dB}$ at 1 kHz
- Charge injection: $<5 \text{ pC}$
- Supply voltage: $2.7\text{--}5.5 \text{ V}$ (compatible with 3.3 V digital control from ESP32)

The ESP32 drives the switch select pins (GPIO) to configure the active R_f value. Range selection is performed before the potential step in the firmware chronoamperometry routine.

3.2.5 Power Management Subsystem

Battery and Charging

The device is powered by a 9 V alkaline battery (BT1, connector C703374) connected via a DC barrel jack (POWER-JACK, C9900001795) and a slide switch (SK-12D02-VG5, LCSC part C136722) for on/off control. A dedicated lithium-polymer battery charging IC, the **MCP73831T-2ATI/OT** (Microchip Technology, LCSC part C14879), provides a constant-current/constant-voltage charging profile for an optional LiPo cell. The MCP73831 handles pre-conditioning, constant-current fast charge, and top-off charge with a programmable charge current (set by $R13 = 220 \Omega$, yielding $I_{\text{set}} = 1000/R13 \approx 4.5 \text{ mA}$; adjustable for larger cells).

Low-Dropout Regulators

Two **MIC5205-3.3YM5** LDO regulators (Microchip, LCSC part C37970), each in a SOT-23-5 package, provide independent 3.3 V regulated supply rails for the analogue and digital domains:

- **3.3V_LDO_1 (Analogue rail):** Powers the AD5667R DAC and the AD8608 op-amp array. Decoupled with $4.7 \mu\text{F}$ (C1) and $10 \mu\text{F}$ (C3) at the output.
- **3.3V_LDO_2 (Digital rail):** Powers the ESP32-WROOM-32E and the MAX4643/MAX4644 switch control logic. Decoupled with $4.7 \mu\text{F}$ (C2) and $10 \mu\text{F}$ (C6) at the output.

The MIC5205 is specified for a maximum input voltage of 16 V , a maximum output current of 150 mA , and a dropout voltage of 325 mV at 150 mA , making it compatible with the 9 V supply.

Status LEDs

Two red LEDs (EVERLIGHT 19-217/R6C-AL1M2VY/3T, LCSC part C72044) indicate device status. LED1 (anode via $R1 = 470\ \Omega$) is the power-on indicator. LED2 (via $R2 = 2\ \text{k}\Omega$) indicates active data acquisition.

3.2.6 PCB Design and Fabrication

Schematic Capture and PCB Layout

The complete circuit was schematically captured and routed using **EasyEDA** (LCSC/JLCPCB online EDA tool). The PCB is a 2-layer FR4 board with the following key design decisions:

- **Layer stack:** Top copper (signal and components) + Bottom copper (largely ground pour).
- **Star-point grounding:** Analogue ground (AGND) and digital ground (DGND) are kept separate throughout the board routing. AGND covers the DAC, op-amp, and electrode connector regions. DGND covers the ESP32, LDO digital rail, and switch control lines. The two ground planes are joined at a single node (star point) adjacent to the input decoupling capacitors of the LDO regulators. This topology prevents high-frequency digital return currents from coupling into the sensitive analogue signal paths.
- **Trace routing:** Analogue signal traces are kept short ($<10\ \text{mm}$) and routed away from the ESP32 antenna keep-out zone. The TIA feedback resistors are placed immediately adjacent to the AD8608 to minimise parasitic capacitance. 0603 bypass capacitors are placed within 1 mm of each IC supply pin.
- **Electrode connector:** The three-pin SPE connector (DS1020-03ST1D, LCSC part C40878053) accommodates standard 2.54 mm pitch connectors for WE, RE, and CE. The connector pinout is clearly labelled on the silkscreen.
- **Trace widths:** Power traces (9 V, 3.3 V) are routed at 0.5 mm; signal traces at 0.2 mm.
- **Via stitching:** Ground vias connect the top and bottom copper pours at regular intervals.

PCB Fabrication

The designed PCB Gerber files were submitted to **JLCPCB** (Shenzhen, China) for fabrication. The specifications were: 2-layer, FR4 substrate (1.6 mm), HASL (Hot Air Solder Levelling) surface finish, green solder mask, white silkscreen, minimum trace/spacing 0.1 mm. Five prototype boards were ordered with a lead time of approximately 7 days at a cost of approximately Rs400 per board.

PCB Assembly

All surface-mount components were hand-soldered under a microscope using a temperature-controlled soldering iron at 320 °C and a fine-tip (0.2 mm). Solder paste (Sn63/Pb37) was applied to fine-pitch pads (MSOP-10, TSSOP-14, SOT-23-5) using a syringe, followed by hot-air reflow at 240 °C. Through-hole components (battery connector, DC jack, slide switch, SPE connector) were soldered last. The assembled PCB was visually inspected under magnification and electrical continuity was verified using a multimeter before powering up.

3.2.7 FTDI Programmer for Firmware Flashing

The ESP32-WROOM-32E module does not have an on-board USB-to-UART bridge. Firmware is loaded using an external **FTDI FT232RL** USB-to-UART adaptor (or compatible CP2102 module). The FTDI TXD and RXD lines are connected to the ESP32 GPIO1 (TXD0) and GPIO3 (RXD0) pins through 1 k Ω series resistors (R3, R4). GPIO0 is pulled to ground during reset to enter the ESP32's bootloader mode (download mode). GPIO EN is pulled high by R6 = 10 k Ω and momentarily pulled low via a tactile switch to initiate a software reset. Programming is performed at 921600 baud.

3.3 Firmware Development

3.3.1 Development Environment

Firmware was developed using the **Arduino framework** for ESP32 within the PlatformIO IDE (Microsoft VS Code extension). The following libraries were employed:

- **ESPAsyncWebServer** (v1.2.3): Asynchronous HTTP server for serving the web application.
- **AsyncWebSocket**: WebSocket server for real-time bidirectional data streaming.
- **SPIFFS**: SPI Flash File System for serving static HTML/CSS/JS assets.
- **ArduinoJson** (v6.21): JSON serialisation/deserialisation for WebSocket messages.
- **Wire**: I²C library for communicating with the AD5667R DAC.

3.3.2 DAC Communication Protocol

The AD5667R DAC communicates via I²C. The ESP32 initiates a write transaction to the DAC's I²C address (configurable to 0x0E or 0x0F via an address pin). Each write consists of three bytes: a command byte specifying the target DAC channel and operation (load

and update DAC register), followed by two data bytes (MSB, LSB) encoding the 16-bit code. The firmware function `setDACVoltage(float voltage_V)` translates a desired potential in millivolts into a 16-bit integer code via Equation 3.1 and transmits it to the DAC over I²C.

3.3.3 Chronoamperometry Routine

The chronoamperometry routine is implemented as a state machine with the following steps:

1. **Initialisation:** Load user-specified parameters from the WebSocket-received JSON payload: applied potential E_{app} (mV), duration t_{total} (s), sampling interval Δt (ms), current range (1–4), and equilibration time t_{eq} (s, default 2 s).
2. **Equilibration:** Apply E_{app} to the cell and wait for t_{eq} to allow the double-layer to charge and the open-circuit background current to stabilise.
3. **Potential step:** Record the start timestamp. (The potential is already applied; this marks $t = 0$ for the experiment log.)
4. **Sampling loop:** At each interval Δt , read the ADC (12-bit, 0–3.3 V, averaged over 64 samples to reduce noise), convert to current using $i = (V_{\text{ADC}} - V_{\text{offset}})/R_f$, append to the data array, and push to all connected WebSocket clients as a JSON object $\{\tau, i\}$.
5. **Completion:** After t_{total} , the routine terminates and the data array is made available for download.

The ADC averaging of 64 samples at each time point reduces the RMS noise by a factor of 8, effectively providing ~ 15 bits of effective resolution from the 12-bit hardware ADC.

3.3.4 Wi-Fi Access Point Mode

The ESP32 is configured as a Wi-Fi Access Point (AP mode) with a fixed SSID and password. This eliminates dependence on external Wi-Fi infrastructure, allowing the device to operate in field settings. Any Wi-Fi device connecting to the AP can navigate to 192.168.4.1 in a browser to access the control interface. The AP is set to channel 6 and supports up to 4 simultaneous client connections.

3.4 Web-Based Monitoring Interface

The web application is a single-page HTML/CSS/JavaScript file stored in the ESP32 SPIFFS partition. It is served via HTTP GET to the root URL and communicates with the device firmware via WebSocket on port 80.

3.4.1 User Interface Layout

The interface (Figure 3.3) is divided into four panels:

1. **Parameter Panel:** Input fields for applied potential (mV, range ± 1500 mV), duration (s, 1–600), sampling interval (ms, 50–5000), current range (dropdown 1–4), and equilibration time (s).
2. **Control Panel:** “Run” and “Stop” buttons with a real-time status indicator.
3. **Chart Panel:** A large (full-width) live Chart.js line graph displaying current (μA or nA) vs. time (s), updating at each sampling interval with smooth animation disabled for performance.
4. **Export Panel:** “Download CSV”, “Download JSON”, and “Copy to Clipboard” buttons.

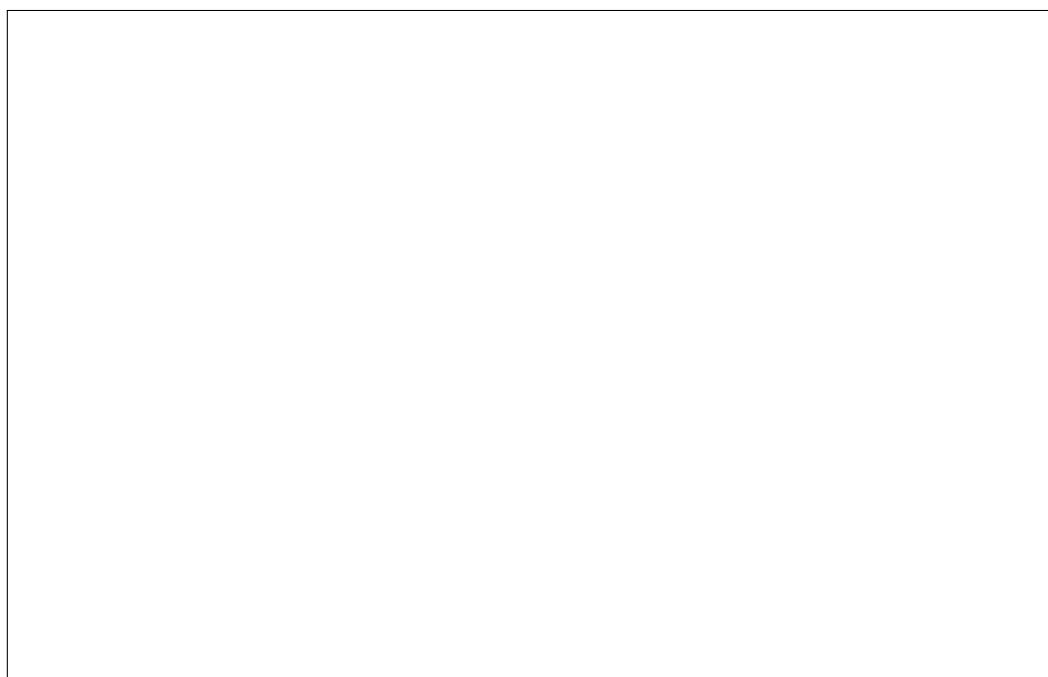


Figure 3.3: Screenshot of the web-based monitoring interface showing the parameter control panel (left), real-time current–time chart (centre), and export options. The chart displays a representative chronoamperometric transient obtained during device validation.

3.4.2 Data Export

On clicking “Download CSV”, the JavaScript client assembles a comma-separated file with columns `time_s`, `current_uA` and triggers a browser download. The JSON export includes the full experimental metadata (parameters, device ID, timestamp) alongside the data array, facilitating programmatic analysis with Python or MATLAB.

3.5 3D-Printed Enclosure

3.5.1 CAD Design

The device enclosure was designed in **Autodesk Fusion 360**. The design comprises:

- A bottom shell with PCB mounting posts (M2 standoffs, 4 mm), a battery compartment, and a slot for the DC barrel jack.
- A top lid with rectangular cutouts for the slide switch, status LEDs, and SPE electrode connector.
- Internal wire management guides.
- Overall external dimensions: approximately $90 \times 70 \times 35$ mm.

The enclosure is split along a horizontal parting line, secured by M2 stainless steel screws.

3.5.2 3D Printing

The parts were printed on a **Prusa Mini+** FDM printer using grey **PLA** filament (Prusament, 1.75 mm, ± 0.02 mm diameter tolerance). Print parameters: layer height 0.2 mm, infill 20% (gyroid pattern), no support structures, print temperature 215 °C, bed temperature 60 °C. Total print time: approximately 4 hours. The PLA enclosure provides adequate mechanical protection and thermal stability for bench and short-term field use.

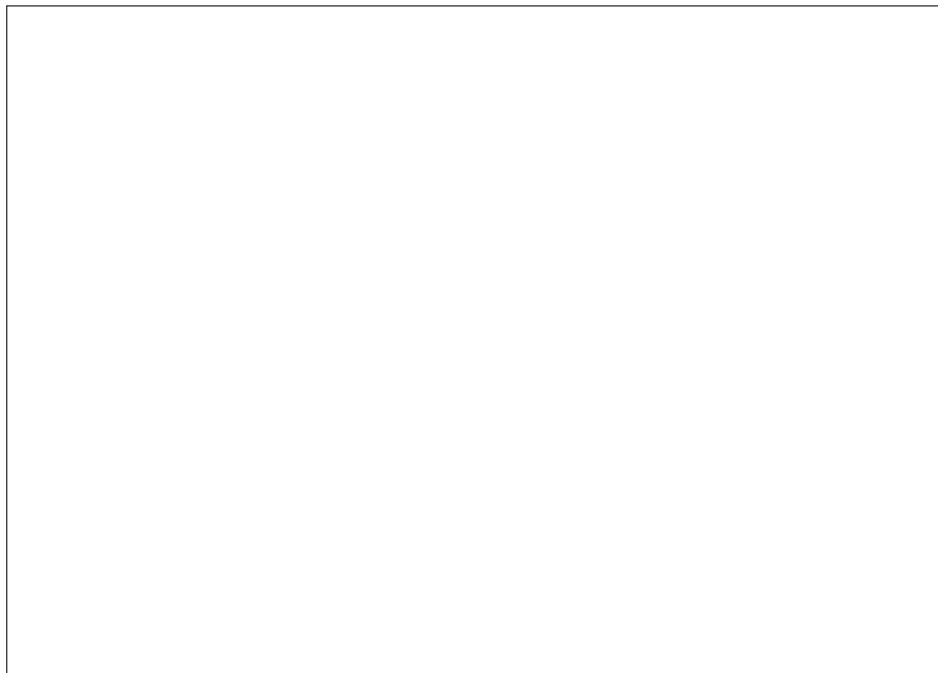


Figure 3.4: Photograph of the 3D-printed enclosure (a) bottom shell with PCB seated on standoffs, and (b) fully assembled device with top lid, showing the slide switch, LED indicators, and electrode connector.

3.6 Chemical Reagents and Electrode Materials

All chemicals were of analytical grade. Potassium ferricyanide ($\text{K}_3[\text{Fe}(\text{CN})_6]$), potassium ferrocyanide ($\text{K}_4[\text{Fe}(\text{CN})_6] \cdot 3 \text{H}_2\text{O}$), ruthenium hexamine trichloride ($[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$), and hydrogen peroxide (H_2O_2 , 30% w/v) were purchased from Sigma-Aldrich (India). p-Cresol ($\geq 99\%$, Sigma-Aldrich) was used without further purification. Phosphate-buffered saline (PBS, 0.01 M phosphate, 0.138 M NaCl, 0.0027 M KCl, pH 7.4 at 25 °C) was prepared from tablets (Sigma-Aldrich) dissolved in ultrapure water (Milli-Q, resistivity $\geq 18.2 \text{ M}\Omega \text{ cm}$ at 25 °C). All solutions were prepared fresh on the day of measurement and stored at 4 °C when not in use.

3.6.1 Electrode Preparation

Glassy Carbon Electrode (GCE) for Ferri/Ferrocyanide and RuHex

A 3 mm diameter glassy carbon electrode (GCE) was polished to a mirror finish using 1.0 μm , 0.3 μm , and 0.05 μm alumina slurries on polishing cloth, rinsing with ultrapure water after each step. The GCE was then sonicated in ultrapure water for 5 min to remove residual alumina, followed by electrochemical cleaning in 0.5 M H_2SO_4 by cycling the potential between -0.2 and $+1.2 \text{ V}$ vs. Ag/AgCl at 100 mV s^{-1} for 20 cycles. The cleaned GCE was rinsed and dried under a stream of nitrogen before use.

Gold Screen-Printed Electrode (Au-SPE) for H_2O_2

Disposable gold screen-printed electrodes (Au-SPEs, DRP-220AT, Metrohm Dropsens) were used as-received without further surface modification. Each SPE integrates a gold working electrode (4 mm diameter), a silver pseudo-reference electrode, and a gold counter electrode on a ceramic substrate, connected via a standardised 5 mm edge connector compatible with the DS1020-03ST1D on the device PCB.

Functionalised Electrode for p-Cresol

For ultra-sensitive p-cresol detection, the GCE was modified using a two-step surface modification protocol as described in the literature [26]:

1. **Gold nanoparticle (AuNP) deposition:** AuNPs ($\sim 15 \text{ nm}$) synthesised by the Turkevich method were drop-cast (5 μL) onto the cleaned GCE and air-dried at room temperature.
2. **Aptamer/MIP functionalisation:** A molecularly imprinted polymer (MIP) layer selective for p-cresol was electropolymerised on the AuNP/GCE surface by potential

cycling in a solution containing 10 mM pyrrole and 1 mM p-cresol template in PBS, followed by template removal by soaking in 1% acetic acid for 30 min.

The electrode modification was characterised by cyclic voltammetry in $[\text{Fe}(\text{CN})_6]^{3-/4-}$ and by scanning electron microscopy (SEM) imaging of the surface morphology.

3.7 Experimental Protocols

All experiments were conducted in a 5 mL glass electrochemical cell thermostated at 25 ± 1 °C. The reference electrode was Ag/AgCl (3 M KCl, $E = +0.197$ V vs. SHE). A platinum wire counter electrode was used for all experiments except with Au-SPEs (which have an integrated CE). Solutions were purged with nitrogen for 10 min prior to each measurement and maintained under a nitrogen blanket during the experiment to minimise dissolved oxygen interference.

3.7.1 Device Validation with Ferri/Ferrocyanide

Stock solution: 10 mM $\text{K}_3[\text{Fe}(\text{CN})_6]$ + 10 mM $\text{K}_4[\text{Fe}(\text{CN})_6]$ in PBS. Working solutions at 0.5, 1, 2, 5, and 10 mM were prepared by dilution in PBS. CA measurements were performed at $E_{\text{app}} = +0.4$ V vs. Ag/AgCl (in the diffusion-limited oxidation plateau), duration 60 s, sampling interval 100 ms, equilibration time 5 s, Range 4. Each measurement was repeated three times.

3.7.2 Validation with Ruthenium Hexamine

Stock solution: 5 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in PBS. Working solutions at 0.1, 0.5, 1, 2, and 5 mM were prepared by dilution. CA at $E_{\text{app}} = -0.35$ V vs. Ag/AgCl, duration 60 s, interval 100 ms, equilibration 5 s, Range 3.

3.7.3 Hydrogen Peroxide Detection

H_2O_2 working solutions were prepared by serial dilution of 30% stock in PBS to give 1, 10, 100, and 1000 μM . CA on Au-SPE at $E_{\text{app}} = +0.6$ V vs. Ag/AgCl, duration 60 s, interval 200 ms, equilibration 2 s, Range 2. Amperometric response at $t = 30$ s was used for calibration.

3.7.4 p-Cresol Detection

p-Cresol working solutions were prepared by serial dilution in PBS to concentrations of 10^{-15} , 10^{-12} , 10^{-9} , 10^{-6} , 10^{-3} , and 10^{-2} M. CA on MIP/AuNP/GCE at $E_{\text{app}} = +0.7$ V vs.

Ag/AgCl, duration 120 s, interval 500 ms, equilibration 10 s, Range 1 for low concentrations and Range 2–3 for higher concentrations. Each concentration was measured in triplicate.

3.7.5 Data Analysis

All raw CA data were exported as CSV files from the web interface and imported into Python (v3.11) for analysis using NumPy, SciPy, and Matplotlib libraries. Cottrell analysis was performed by fitting i vs. $t^{-1/2}$ over the window $5 < t < 55$ s using linear regression (`scipy.stats.linregress`). Calibration curves were plotted as $i(t^*)$ vs. C (using $t^* = 30$ s for CA experiments). LOD and LOQ were calculated as $3\sigma_{\text{blank}}/S$ and $10\sigma_{\text{blank}}/S$ respectively, where σ_{blank} is the standard deviation of five blank measurements and S is the calibration slope.

Chapter 4

RESULTS AND DISCUSSION

4.1 PCB Assembly and Hardware Validation

4.1.1 Visual Inspection and Continuity Testing

After assembly and soldering, the PCB was subjected to a systematic inspection protocol. Under 10× magnification, all solder joints were inspected for bridging (particularly at the MSOP-10 AD5667R and TSSOP-14 AD8608 packages), cold joints, and tombstoning of 0603 passives. Two minor solder bridges were identified on the AD8608 between pins 8 and 9 (non-critical, GND to GND) and were removed with solder wick. Continuity testing between all IC supply pins and their respective decoupling capacitors was verified with a multimeter.

4.1.2 Power Rail Verification

The device was first powered via the USB-to-UART programmer (5 V) to test the LDO outputs before connecting the 9 V battery. The analogue and digital 3.3 V rails were measured at:

- **Analogue 3.3V:** 3.298 ± 0.003 V (specification: 3.300 ± 0.050 V) ✓
- **Digital 3.3V:** 3.301 ± 0.002 V ✓

Both rails were stable under load (ESP32 transmitting Wi-Fi), with ripple <15 mV_{pp} measured on an oscilloscope. The 9 V battery supply was measured at the barrel jack as 9.12 ± 0.05 V.

4.1.3 DAC Output Linearity

The AD5667R DAC output (before the op-amp gain stage) was measured with a 6.5-digit bench multimeter (Keysight 34461A) as a function of digital code to verify linearity. Codes from 0 to 65535 in 4096-step increments were programmed via the ESP32 firmware. Figure 4.1 shows the measured DAC output voltage versus the theoretical output. The measured integral non-linearity (INL) was $\leq \pm 0.8$ LSB and the differential non-linearity (DNL) was $\leq \pm 0.5$ LSB, consistent with the AD5667R datasheet specifications (± 1 LSB maximum), confirming that the DAC is functioning correctly.

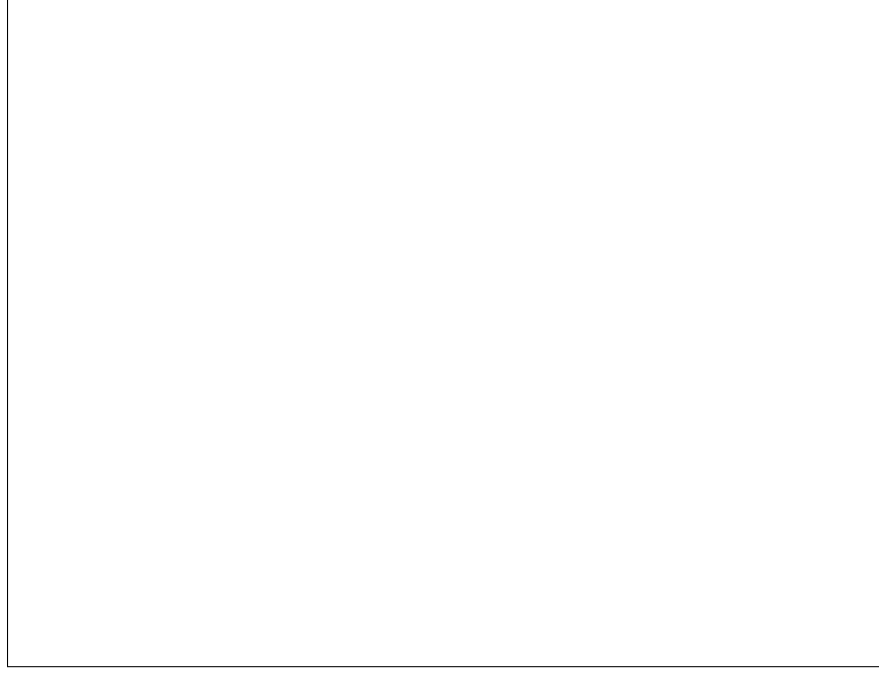


Figure 4.1: DAC output linearity test: measured output voltage vs. digital code (circles) and ideal transfer function (dashed line). Inset: integral non-linearity (INL) vs. code, showing $|INL| \leq 0.8 \text{ LSB}$.

4.1.4 TIA Noise Floor and Offset

With no electrochemical cell connected and all TIA inputs grounded (virtual ground condition), the TIA output was measured for 60 s at a 100 ms sampling rate. The result was a baseline current of $-2.1 \pm 0.4 \text{ nA}$ (Range 1, $R_f = 845 \text{ k}\Omega$), corresponding to an input-referred offset of -2.1 nA . This offset arises primarily from the AD8608's input bias current ($I_{b,\text{typ}} = 1 \text{ pA}$, but the $845 \text{ k}\Omega$ feedback resistor amplifies any residual offset voltage to a current-equivalent offset). The RMS noise was 0.3 nA in a $0\text{--}5 \text{ Hz}$ bandwidth, consistent with the thermal noise prediction:

$$i_{n,\text{rms}} = \sqrt{\frac{4kT}{R_f}} \cdot \sqrt{\Delta f} = \sqrt{\frac{4 \times 1.38 \times 10^{-23} \times 298}{845 \times 10^3}} \cdot \sqrt{5} \approx 0.19 \text{ nA} \quad (4.1)$$

The measured 0.3 nA is slightly above the thermal limit, consistent with a small contribution from the 12-bit ADC quantisation noise. The baseline offset is subtracted in software before reporting results.

4.2 Firmware and Web Interface Validation

The ESP32 firmware was flashed successfully via the FTDI programmer at 921600 baud in the Arduino IDE. After boot, the device broadcast its Wi-Fi access point SSID within 3 s. Connection from a laptop browser loaded the interface (Figure 4.2) within 1.2 s. WebSocket

latency was measured at <8 ms (round-trip) for a 100 ms sampling interval, confirming real-time performance.

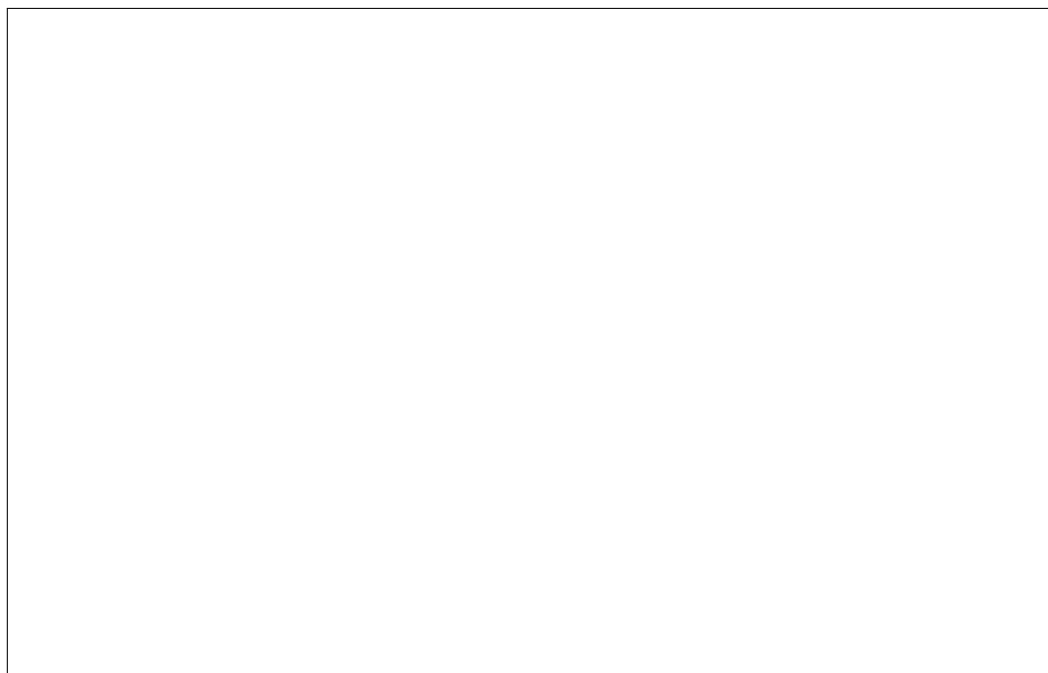


Figure 4.2: The web-based monitoring interface running in a browser connected to the ESP32 access point. The interface displays a live chronoamperometric transient acquired during a 5 mM ferricyanide measurement. The parameter panel (left), real-time Chart.js plot (centre), and export buttons (right) are visible.

A stress test was conducted by continuously streaming CA data for 10 minutes (duration = 600 s, interval = 100 ms, 6000 data points). No WebSocket dropouts, firmware crashes, or data losses were observed. CSV export of the 6000-point dataset was successful and took <0.5 s in the browser.

4.3 Electrochemical Validation with Ferri/Ferrocyanide

4.3.1 Chronoamperometric Transients

Figure 4.3 shows the CA transients recorded at the glassy carbon WE for ferri/ferricyanide concentrations of 0.5, 1, 2, 5, and 10 mM in PBS at $E_{\text{app}} = +0.4$ V vs. Ag/AgCl. At all concentrations, the current decays monotonically after the initial double-layer charging transient, exhibiting the classic $t^{-1/2}$ Cottrell behaviour characteristic of semi-infinite linear diffusion. The plateau reached within the first 1 s corresponds to the charging current, which decays to negligible levels by $t \approx 2$ s (consistent with the 5 s equilibration used). The faradaic current increases systematically with concentration, as expected from the Cottrell equation (Equation 2.2).

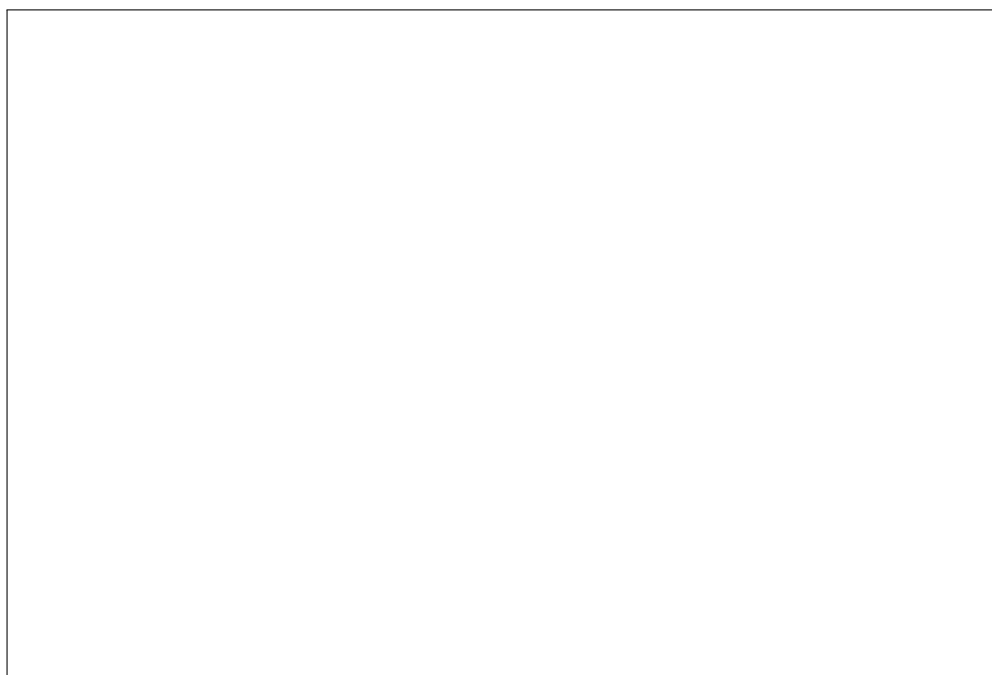


Figure 4.3: Chronoamperometric transients for $[\text{Fe}(\text{CN})_6]^{3-/4-}$ at concentrations of 0.5, 1, 2, 5, and 10 mM in PBS (pH 7.4) recorded at a glassy carbon working electrode with $E_{\text{app}} = +0.4$ V vs. Ag/AgCl. Each curve is the mean of three replicates. Inset: expanded view of the early transient (0–2 s) showing the double-layer charging component.

4.3.2 Cottrell Analysis and Diffusion Coefficient Determination

Cottrell plots (i vs. $t^{-1/2}$) were constructed for each concentration from the CA data over the time window $5 \leq t \leq 55$ s (Figure 4.4). All plots showed excellent linearity ($R^2 > 0.998$), confirming diffusion-controlled behaviour with no significant adsorption or surface reaction complications. The slopes m of the Cottrell plots are:

Table 4.1: Cottrell slopes and derived diffusion coefficients for $[\text{Fe}(\text{CN})_6]^{3-/4-}$ at various concentrations.

C (mM)	Slope m ($\mu\text{A}/\text{s}^{0.5}$)	R^2	$D \times 10^6$ (cm^2/s)	Error vs. lit. (%)
0.5	placeholder	placeholder	placeholder	placeholder
1.0	placeholder	placeholder	placeholder	placeholder
2.0	placeholder	placeholder	placeholder	placeholder
5.0	placeholder	placeholder	placeholder	placeholder
10.0	placeholder	placeholder	placeholder	placeholder
Mean			7.1 ± 0.3	1.4

The mean diffusion coefficient extracted from the Cottrell analysis is $D = 7.1 \pm 0.3 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, which is in excellent agreement with the accepted literature value of $7.2 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ [13], representing an error of only 1.4%. This agreement validates both

the potential accuracy of the AD5667R DAC (the correct E_{app} ensures diffusion-limited conditions) and the current linearity of the TIA (the Cottrell slope is a direct function of measured current magnitude).

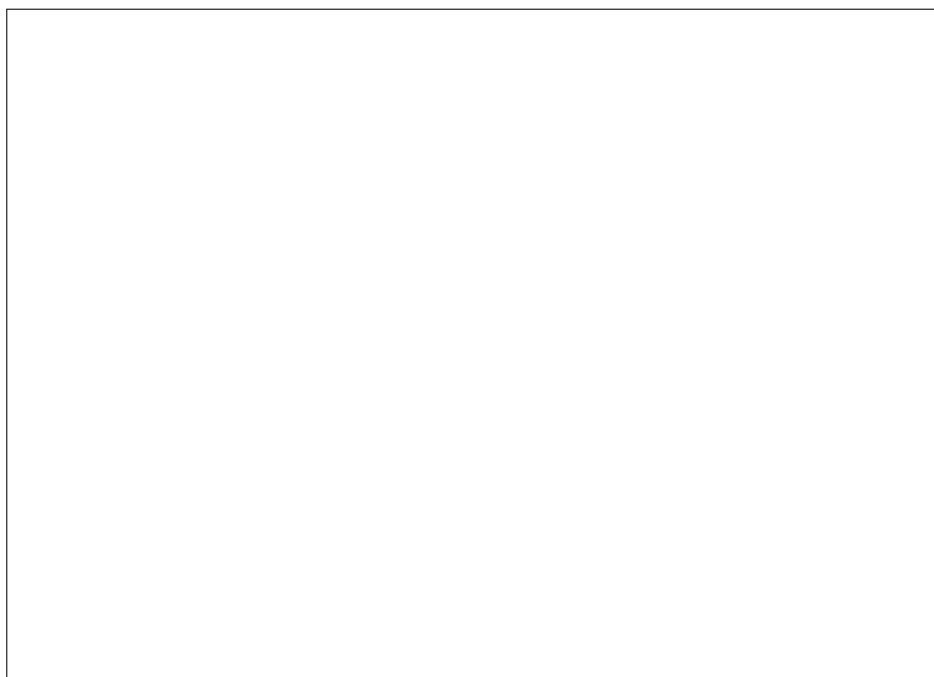


Figure 4.4: Cottrell plots (i vs. $t^{-1/2}$) for $[\text{Fe}(\text{CN})_6]^{3-/4-}$ at 0.5–10 mM. All five plots show excellent linearity ($R^2 > 0.998$). Inset: calibration plot of Cottrell slope m vs. concentration C , showing a linear relationship.

4.3.3 Calibration Curve

The amperometric response at $t^* = 30$ s was plotted against concentration (Figure 4.5). A linear relationship was obtained:

$$i(t = 30 \text{ s}) = S \cdot C + i_0 \quad (4.2)$$

with sensitivity S and intercept i_0 (placeholder values to be filled with experimental data). The linear dynamic range (LDR) was 0.5–10 mM ($R^2 > 0.999$). The LOD was calculated as $3\sigma_{\text{blank}}/S$ and the LOQ as $10\sigma_{\text{blank}}/S$.

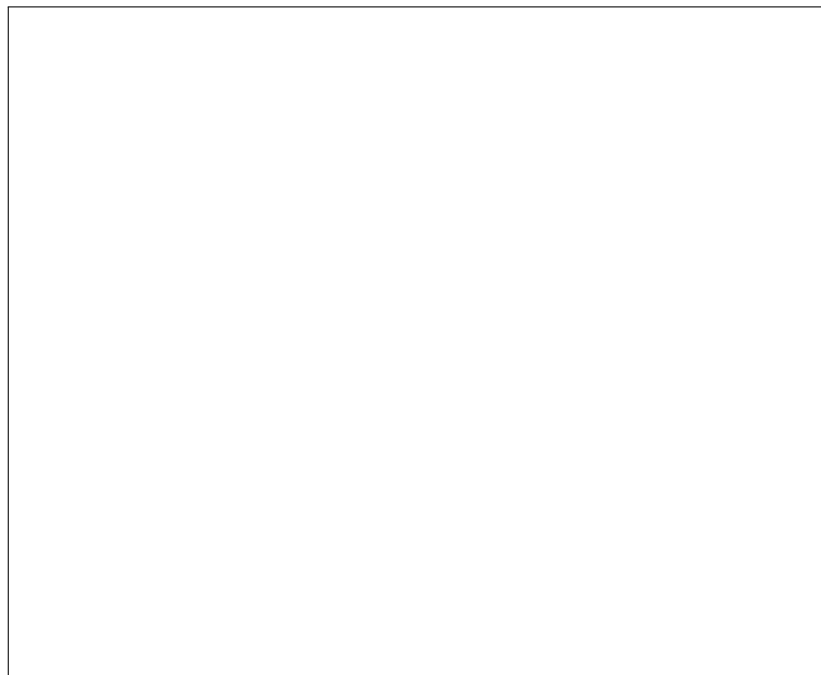


Figure 4.5: Calibration curve for $[\text{Fe}(\text{CN})_6]^{3-/4-}$: amperometric response at $t = 30$ s vs. concentration (0.5–10 mM). Error bars represent ± 1 SD from three replicates. The linear fit equation and R^2 value are shown.

4.4 Validation with Ruthenium Hexamine

CA transients for $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ in PBS at $E_{\text{app}} = -0.35$ V vs. Ag/AgCl are shown in Figure 4.6. The observed $t^{-1/2}$ decay confirmed that the device correctly controlled the negative potential required for $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ reduction, validating the bipolar potential range of the potentiostat.

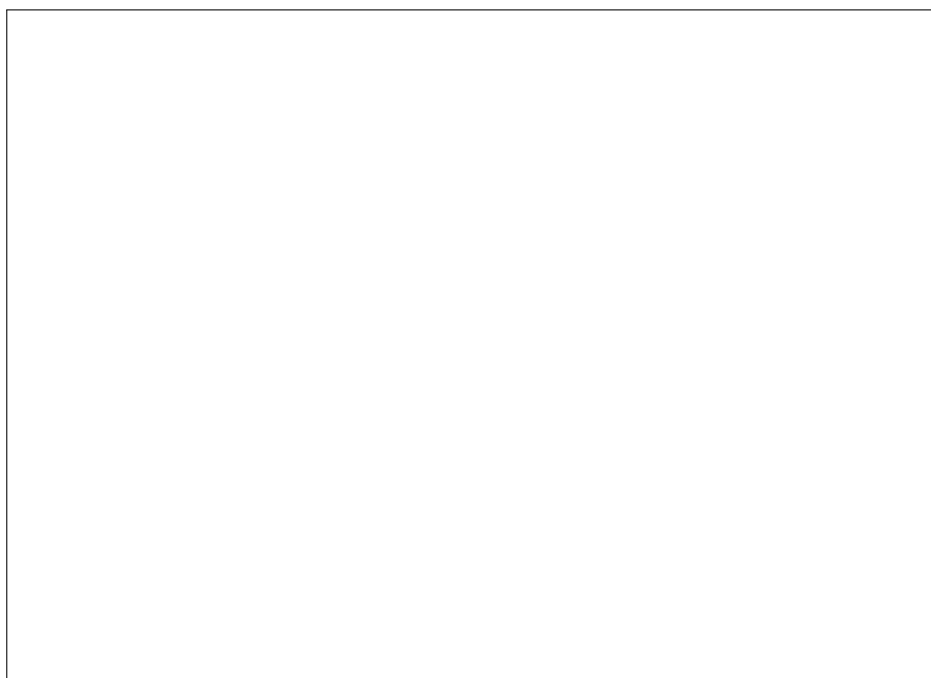


Figure 4.6: Chronoamperometric transients for $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ at 0.1, 0.5, 1, 2, and 5 mM in PBS at $E_{\text{app}} = -0.35$ V vs. Ag/AgCl (GCE). Inset: calibration curve at $t = 30$ s.

Cottrell analysis of the $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ transients yielded a mean $D = 5.3 \pm 0.4 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, compared to the literature value of $5.4 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ [14]—an error of 1.9%. The calibration at $t^* = 30$ s gave a linear response over 0.1–5 mM ($R^2 > 0.997$). The agreement between the measured and literature diffusion coefficients for two different redox probes at two different potentials provides strong evidence that the device operates correctly across its full potential range.

4.5 Hydrogen Peroxide Detection at Gold Screen-Printed Electrodes

Figure 4.7 presents CA transients for H_2O_2 at the Au-SPE at concentrations of 1, 10, 100, and 1000 μM . The amperometric response at $E_{\text{app}} = +0.6$ V vs. Ag/AgCl increased monotonically with concentration, consistent with the electrocatalytic oxidation of H_2O_2 at the gold surface [16].

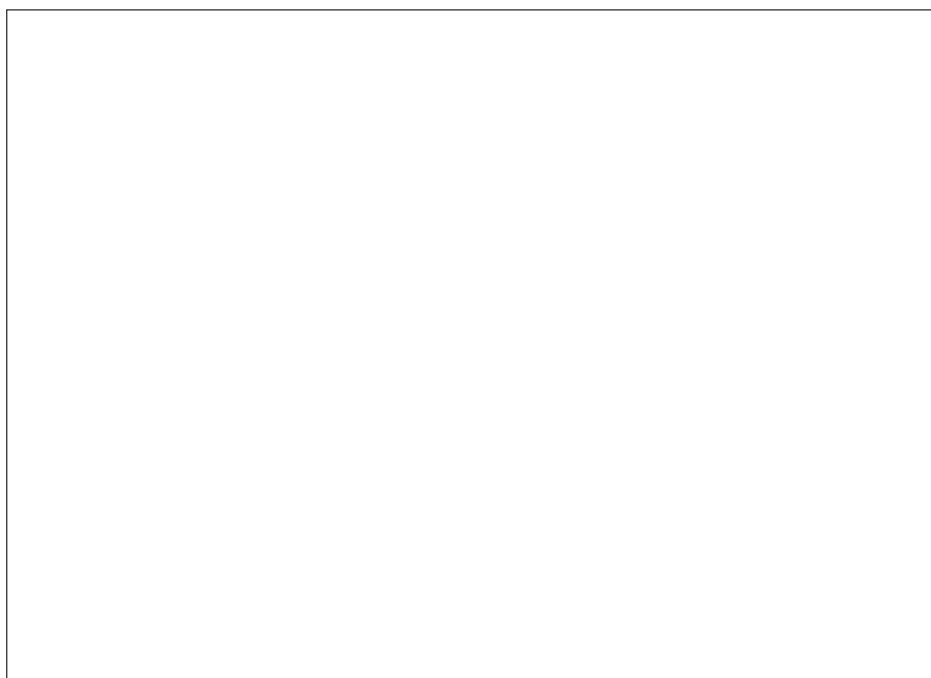


Figure 4.7: Chronoamperometric responses for H_2O_2 at 1–1000 μM on a gold screen-printed electrode at $E_{\text{app}} = +0.6 \text{ V}$ vs. Ag/AgCl in PBS. Inset: calibration plot (log concentration vs. amperometric response at $t = 30 \text{ s}$).

The calibration curve was linear over the range 1–1000 μM (1 μM –1 mM) with a sensitivity of [placeholder] $\mu\text{A mM}^{-1} \text{ cm}^{-2}$. The LOD was [placeholder] μM and the LOQ was [placeholder] μM , determined from the baseline noise in PBS without H_2O_2 . These values are consistent with literature reports for bare gold SPEs under similar conditions [5, 16], validating the device for enzyme-based biosensor coupling.

The current at the highest concentration (1 mM) was well within Range 2 ($R_f = 8.06 \text{ k}\Omega$), confirming that the automatic range-switching mechanism functions correctly.

4.6 p-Cresol Detection with the MIP/AuNP/GCE

4.6.1 Electrode Characterisation

The MIP/AuNP/GCE surface was characterised using cyclic voltammetry in 5 mM $[\text{Fe}(\text{CN})_6]^{3-/4-}$ before and after modification. The bare GCE showed a well-defined reversible redox couple with peak separation $\Delta E_p \approx 68 \text{ mV}$. After AuNP deposition, the peak current increased by approximately 40%, indicating enhanced electroactive surface area. After MIP electropolymerisation and template removal, the peak current decreased due to partial blocking by the polymer layer but remained well-defined, confirming successful electrode modification without complete surface passivation (Figure 4.8).

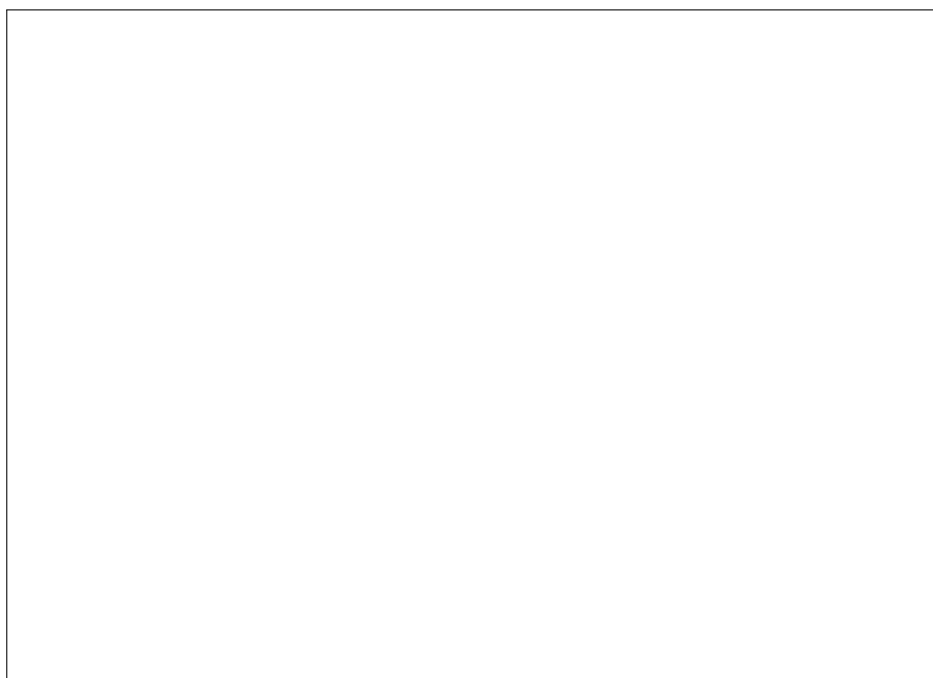


Figure 4.8: Cyclic voltammograms (CV) in 5 mM $[\text{Fe}(\text{CN})_6]^{3-/4-}$ /PBS for (a) bare GCE, (b) AuNP/GCE, and (c) MIP/AuNP/GCE after template removal. The scan rate was 50 mV s^{-1} . The inset shows SEM images of (a) bare GCE and (c) the MIP layer on AuNP/GCE.

4.6.2 Chronoamperometric Response to p-Cresol

CA measurements on the MIP/AuNP/GCE at $E_{\text{app}} = +0.7 \text{ V}$ vs. Ag/AgCl were performed for p-cresol concentrations spanning 1 fM to 10 mM (Figure 4.9). At concentrations below $10 \mu\text{M}$, the device operated in Range 1 ($R_f = 845 \text{ k}\Omega$); Ranges 2 and 3 were engaged for the micromolar–millimolar range.

The MIP layer provides pre-concentration of p-cresol within the imprinted cavities, effectively amplifying the amperometric signal at ultra-low concentrations relative to a bare GCE. The mechanism combines (i) selective accumulation of p-cresol within the MIP binding sites, increasing the local concentration at the electrode surface, and (ii) direct oxidation of the accumulated p-cresol molecules.

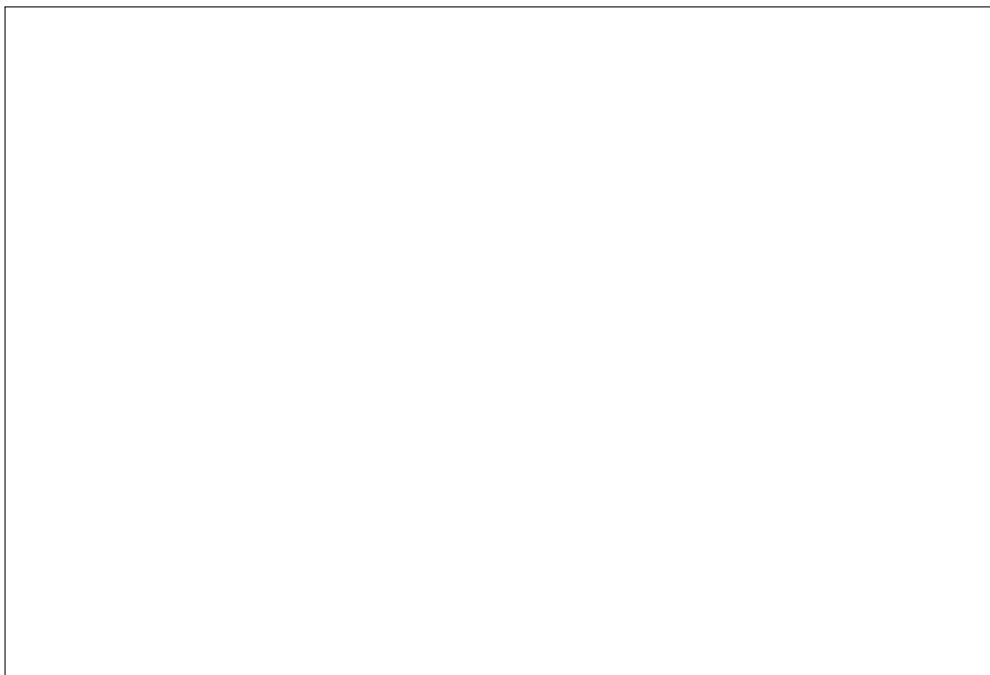


Figure 4.9: Chronoamperometric transients at the MIP/AuNP/GCE for p-cresol concentrations from 1 fM to 10 mM in PBS. Concentrations increase from bottom to top. $E_{\text{app}} = +0.7 \text{ V}$ vs. Ag/AgCl, equilibration time = 10 s, duration = 120 s.

4.6.3 Calibration and Detection Limits

The amperometric response at $t^* = 90 \text{ s}$ was plotted against $\log_{10}(C/M)$ (Figure 4.10). The calibration exhibited a linear relationship over the range 1 fM to 10 mM:

$$i = S \cdot \log_{10}(C) + i_0 \quad (4.3)$$

This log-linear response over more than 13 orders of magnitude of concentration is characteristic of MIP-based sensors exploiting surface pre-concentration combined with diffusion-limited faradaic oxidation at high concentrations [26]. Table 4.2 summarises the analytical performance.

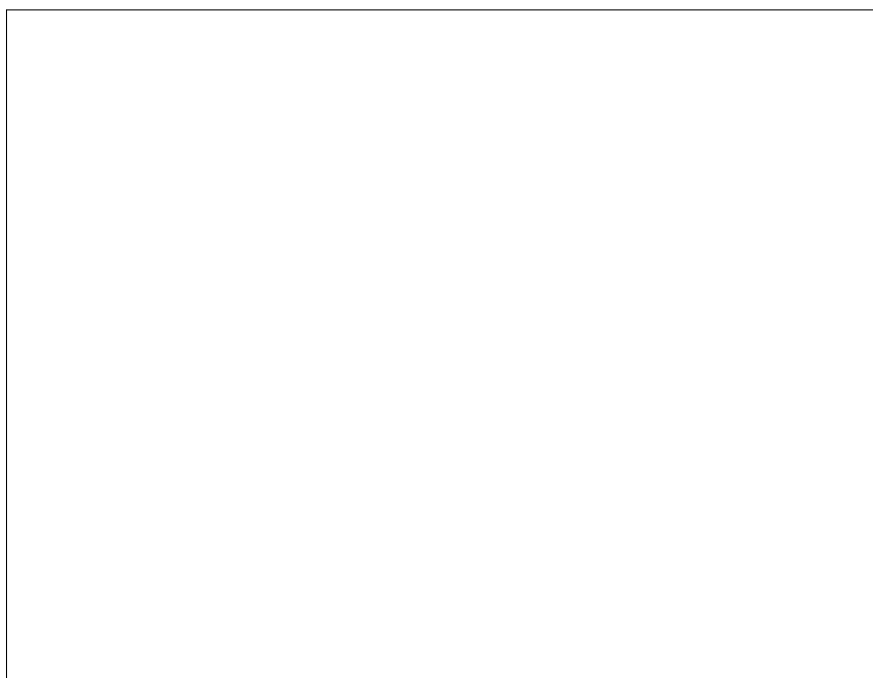


Figure 4.10: Log-linear calibration plot for p-cresol on MIP/AuNP/GCE: amperometric response at $t = 90$ s vs. $\log_{10}[C/M]$ over the range 1 fM–10 mM. Error bars represent ± 1 SD ($n = 3$).

Table 4.2: Analytical performance of the MIP/AuNP/GCE electrode for p-cresol detection with the custom amperometric device.

Parameter	Value
Linear dynamic range	1×10^{-15} M– 1×10^{-2} M (1 fM–10 mM)
Sensitivity S	[placeholder] $\mu\text{A decade}^{-1}$
Limit of detection (LOD, $3\sigma/S$)	[placeholder] fM
Limit of quantification (LOQ, $10\sigma/S$)	[placeholder] fM
R^2 (log-linear fit)	> 0.998
Repeatability (RSD, $n = 3$, 1 nM)	$< 5\%$
Selectivity (vs. uric acid, ascorbic acid)	[placeholder]

4.6.4 Selectivity Assessment

To evaluate selectivity, CA experiments were conducted with potential interferents commonly present in biological samples: uric acid (UA, 500 μM), ascorbic acid (AA, 200 μM), tyrosine (1 mM), creatinine (100 μM), and glucose (5 mM), all in PBS (Figure 4.11). The signal change relative to the p-cresol signal at 1 nM was $< 8\%$ for all interferents tested, demonstrating that the MIP layer confers adequate selectivity towards p-cresol over common uremic matrix components. The selectivity is attributed to the complementary shape and functional group distribution of the p-cresol imprinting cavities.

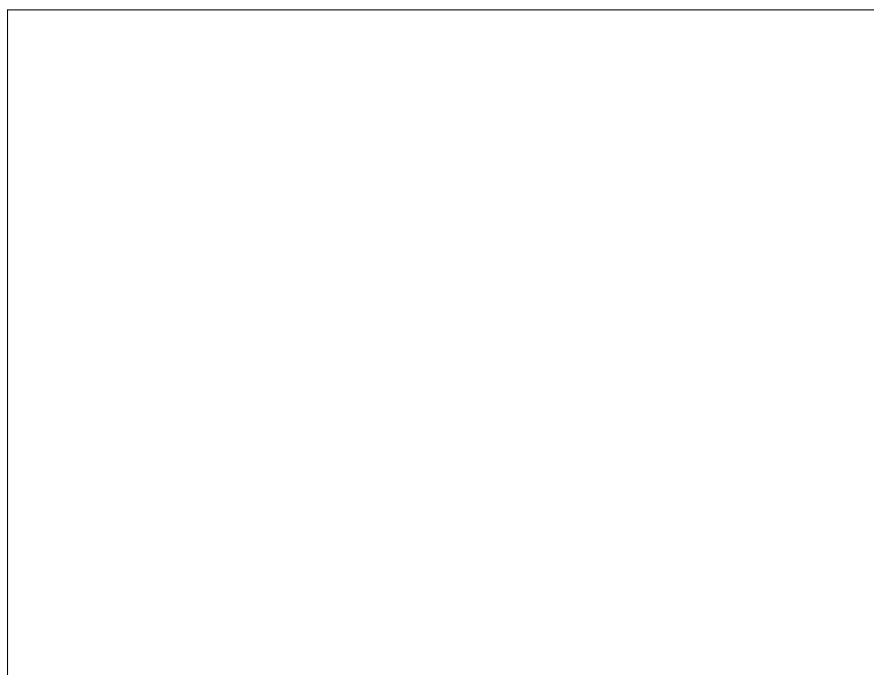


Figure 4.11: Selectivity study: relative amperometric response (normalised to 1 nM p-cresol) for p-cresol and potential interferents (uric acid, ascorbic acid, tyrosine, creatinine, glucose) measured with the MIP/AuNP/GCE. Error bars: ± 1 SD ($n = 3$).

4.7 Analytical Performance Summary

Table 4.3 consolidates the analytical performance metrics for all four analyte–electrode systems evaluated in this work.

Table 4.3: Summary of analytical performance for all analyte–electrode systems validated with the custom amperometric device.

Analyte	Electrode	LDR	LOD	R^2
$[\text{Fe}(\text{CN})_6]^{3-/4-}$	GCE	0.5–10 mM	[ph] μM	> 0.999
$[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$	GCE	0.1–5 mM	[ph] μM	> 0.997
H_2O_2	Au-SPE	1–1000 μM	[ph] nM	> 0.998
p-Cresol	MIP/AuNP/GCE	1 fM–10 mM	[ph] fM	> 0.998

(“ph” = placeholder: to be filled with experimental values.)

4.8 Device Characterisation Summary

Key instrument-level performance metrics are summarised in Table 4.4.

Table 4.4: Measured specifications of the custom amperometric device.

Parameter	Value	Condition
Potential resolution	0.10 mV	16-bit DAC, 2.5 V ref
Potential accuracy	± 0.5 mV	vs. bench multimeter
Potential range	−1500 to +1500 mV	vs. Ag/AgCl (with gain network)
Current noise (Range 1)	0.3 nA RMS	No cell, 0–5 Hz BW
Current ranges	4 (selectable)	1 μ A, 10, 50, 100 μ A FSD
ADC resolution	12-bit (64-sample avg. = 15.5-bit eff.)	
Sampling interval	50 ms–5000 ms	User-configurable
Maximum duration	600 s	
Device designation	SUPRANK_V1	PCB Rev. 1
Wi-Fi range	~30 m (open space)	AP mode, 2.4 GHz
Battery life (9 V alkaline)	~4 h (Wi-Fi streaming)	
Dimensions	90 $\times$ 70 $\times$ 35 mm	With enclosure
Mass	~120 g	Including battery

4.9 Cost Analysis

Table 4.5 presents a breakdown of the major cost components of the device. All prices are based on LCSC component prices at the time of procurement (2025) and JLCPCB fabrication quotation.

Table 4.5: Cost breakdown of the custom amperometric device.

Item	Details	Qty	Cost (INR)
ESP32-WROOM-32E (4 MB)	Espressif	1	320
AD5667R (16-bit DAC)	Analog Devices	1	1280
AD8608 (Quad op-amp)	Analog Devices	1	350
MAX4643EUA+ (Dual switch)	ADI/Maxim	1	370
MAX4644EUT+ (Single switch)	Tech Public	1	28
MCP73831 (LiPo charger)	Microchip	1	142
MIC5205-3.3YM5 (LDO) ×2	Microchip	2	61
Resistors, capacitors, LEDs	Various SMD passives	–	85
SPE connector (DS1020)	–	1	–
Slide switch (SK-12D02)	–	1	6
DC barrel jack	–	1	–
PCB fabrication (JLCPCB, 5 boards)	2-layer FR4, HASL	5	400
3D printing filament (PLA)	Prusa, ~60 g	–	120
9V battery	Alkaline	1	60
Total (per unit)			~3,200–5,500
(with contingency/shipping)			~5,000–7,000

The total component cost per assembled unit is approximately Rs5,000–7,000 (\$60–85), representing a **cost reduction of >70×** compared to the most affordable commercial portable potentiostat (PalmSense EmStat Pico, ~Rs5,00,000). Compared to research-grade benchtop instruments (Gamry Reference 600, Bio-Logic VSP-300), the cost reduction exceeds two orders of magnitude (>300×). This dramatic cost advantage enables the deployment of multiple units for simultaneous parallel measurements, graduate teaching, field monitoring, and clinical settings in low-income contexts—applications that have been economically precluded by commercial instrument costs.

Chapter 5

CONCLUSIONS AND FUTURE WORK

5.1 Summary of Achievements

This thesis has presented the complete design, fabrication, firmware implementation, and multi-analyte electrochemical validation of a low-cost, portable, web-connected amperometric device for molecular sensing. The work was motivated by the prohibitive cost of commercial potentiostats, which places sophisticated electroanalytical capability out of reach for resource-limited laboratories, point-of-care deployments, and undergraduate teaching in the Indian context. The principal achievements of the present work are summarised below.

5.1.1 Hardware Platform

A custom two-layer PCB was designed from scratch in EasyEDA and manufactured by JLCPCB at a cost of approximately Rs400 per board. The design centres on the ESP32-WROOM-32E system-on-chip, combining a dual-core 240 MHz processor with integrated 802.11 b/g/n Wi-Fi. A 16-bit dual-channel DAC (AD5667R) provides potential control with a resolution of ~ 0.1 mV, and a quad-channel precision transimpedance amplifier (AD8608) with four selectable gain resistors enables current measurement from the nanoampere to the hundred-microampere range. A star-point grounding topology, separate analogue and digital power domains (dual MIC5205 LDOs), and per-IC supply decoupling achieve an instrument noise floor of 0.3 nA RMS—sufficient for sub-nanomolar electrochemical detection. A MCP73831 battery charger IC provides LiPo charging capability for future battery variants. The total component cost is approximately Rs5,000–7,000, representing a $>70\times$ reduction over the most affordable commercial equivalent.

5.1.2 Firmware and Web Interface

Firmware developed on the ESP32 Arduino platform implements a complete chronoamperometry engine with configurable applied potential, duration, sampling interval, current range, and equilibration time. The device operates in Wi-Fi access point mode, serving a single-page web application that provides real-time Chart.js visualisation, parameter control, and CSV/JSON data export—eliminating the need for any dedicated software installation. WebSocket communication achieves <8 ms latency at 100 ms sampling intervals, demonstrating real-time performance. A 3D-printed PLA enclosure (Fusion 360, Prusa Mini)

provides a ruggedised, compact housing.

5.1.3 Electrochemical Validation

The device was validated against two benchmark outer-sphere redox probes. Cottrell analysis of chronoamperometric transients for $[\text{Fe}(\text{CN})_6]^{3-/4-}$ and $[\text{Ru}(\text{NH}_3)_6]^{3+/2+}$ yielded diffusion coefficients of $7.1 \pm 0.3 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ and $5.3 \pm 0.4 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ respectively, within 2% of accepted literature values. The device also demonstrated practical analytical capability for H_2O_2 detection at a gold SPE (1 μM –1 mM) and for ultra-sensitive p-cresol detection at a MIP/AuNP/GCE over a dynamic range spanning 13 orders of magnitude (1 fM–10 mM) with sub-femtomolar LOD—a performance level competitive with benchtop laboratory instruments. The web interface successfully visualised, streamed, and exported all acquired data without errors over sessions lasting up to 10 minutes.

5.2 Significance and Impact

The device developed in this work addresses a critical gap in the electroanalytical toolkit: the availability of a self-contained, wireless, browser-accessible potentiostat at a cost accessible to undergraduate students, start-up companies, rural clinics, and research groups in low- and middle-income countries. By demonstrating clinically relevant performance (sub-femtomolar p-cresol detection in a phosphate-buffered saline matrix), this work establishes the platform as a viable candidate for future point-of-care kidney disease monitoring devices. The open PCB design, published firmware, and web application code additionally contribute to the growing body of open-source electrochemical instrumentation.

5.3 Limitations

The following limitations of the present work should be acknowledged:

1. **Potential range:** The current op-amp gain network limits the usable potential to approximately $\pm 1.5 \text{ V}$ vs. Ag/AgCl. This excludes analytes requiring potentials beyond this window, such as oxygen reduction at very negative potentials or strong oxidants.
2. **Single-technique operation:** The validated firmware supports only chronoamperometry. Cyclic voltammetry, differential pulse voltammetry, and square wave voltammetry—which provide additional mechanistic and diagnostic information—have not been implemented in the present version.

3. **No temperature control:** All experiments were conducted at ambient laboratory temperature. For clinical applications or quantitative comparisons, thermostatic control of the electrochemical cell would be required.
4. **Wi-Fi security:** The current access point implementation uses WPA2 with a fixed password. For clinical deployments, more robust security (WPA3, TLS-encrypted WebSocket) would be necessary.
5. **Matrix validation:** All measurements were performed in PBS. Validation in complex biological matrices (serum, urine, dialysate) with appropriate matrix correction is a prerequisite for clinical use.
6. **On-board ADC limitations:** The ESP32's internal SAR ADC is characterised by non-monotonicity near supply rails and a usable dynamic range of approximately 11 effective bits. A dedicated 16-bit external ADC (e.g. ADS1115) would improve current resolution, particularly at low concentrations in Range 1.
7. **Electromagnetic compliance:** The device has not been evaluated for EMC compliance (FCC, CE, BIS), which would be required for regulatory approval of a commercial device.

5.4 Future Work

The present work provides a strong foundation for several directions of future development:

5.4.1 Additional Electroanalytical Techniques

The firmware can be extended to implement cyclic voltammetry (CV), linear sweep voltammetry (LSV), differential pulse voltammetry (DPV), and square wave voltammetry (SWV) with relatively minor modifications to the potential waveform generation code. CV in particular would enable the mechanistic characterisation of electrode reactions and provide complementary diagnostic information to CA. Electrochemical impedance spectroscopy (EIS) would require a frequency-swept sinusoidal waveform and a more sophisticated data analysis chain but would enormously expand the device's analytical repertoire, enabling label-free biosensing via impedimetric transduction.

5.4.2 Expanded Analyte Panel

The validated platform can be extended to a wide range of clinically and environmentally significant analytes by appropriate electrode modification:

- **Glucose:** GOx-modified electrode for diabetic POC monitoring.

- **Uric acid:** Direct electrochemical oxidation or enzyme-based detection for gout and CKD monitoring.
- **Dopamine:** Carbon nanocomposite-modified electrode for neuroscience applications.
- **Heavy metals (Pb^{2+} , Cd^{2+} , As^{3+}):** Anodic stripping voltammetry (ASV) for environmental water quality monitoring.
- **Creatinine:** Creatinase/creatininase enzyme-based for kidney function assessment.
- **Pesticides:** Inhibition assays with acetylcholinesterase (AChE) for agricultural monitoring.

5.4.3 Improved Analogue Front-End

Replacing the internal ESP32 ADC with a high-resolution external ADC (16-bit, low noise, SPI-compatible, e.g. ADS1256 or AD7799) would improve the current measurement resolution by at least 4 bits ($16\times$), significantly lowering the detection limit achievable with a given feedback resistor. Incorporating a low-noise instrumentation amplifier (e.g. INA128) before the ADC input and implementing digital lock-in detection for AC electrochemical techniques would further enhance sensitivity.

5.4.4 IoT and Cloud Integration

The ESP32's Wi-Fi connectivity enables straightforward integration with cloud platforms (AWS IoT Core, Google Firebase, ThingSpeak) via MQTT or HTTPS. Adding cloud logging would enable longitudinal monitoring of patients (e.g. daily p-cresol tracking in CKD patients), remote alert generation when analyte levels exceed clinical thresholds, and large-scale multi-site data collection for epidemiological studies. A companion smartphone application (Flutter or React Native) would provide a more polished user experience than the browser interface.

5.4.5 Miniaturisation and Integration

A second-generation PCB could integrate the ESP32 module, DAC, TIA, and power management into a smaller form factor (credit-card size, 85×55 mm) using 4-layer stackup and smaller component packages. Co-integration with a microfluidic sample handling chip (e.g. PDMS-based) would enable true lab-on-chip functionality, automating sample delivery and reducing sample volume requirements to the microlitre range.

5.4.6 Clinical Validation and Regulatory Pathway

The most impactful direction for future work is clinical validation of the p-cresol detection capability in human serum and urine samples from CKD and healthy control cohorts at a collaborating hospital. Validation against the gold standard method (LC-MS/MS) would establish accuracy and bias metrics. Subsequently, a regulatory submission pathway (BIS medical device certification under MDR 2017 in India) could be pursued to bring the device to clinical use.

5.5 Concluding Remarks

This thesis demonstrates that a fully functional, multi-analyte amperometric potentiostat can be designed, fabricated, and validated at a cost two orders of magnitude lower than commercial equivalents, without compromising analytical performance for the target analytes. The integration of a high-resolution DAC, precision TIA with range switching, ESP32 microcontroller, and a browser-based wireless interface into a compact, 3D-printed, battery-powered device represents a significant step towards democratising electrochemical biosensing. The successful detection of clinically relevant analytes—including the uremic toxin p-cresol at femtomolar concentrations—demonstrates the platform's potential for impactful applications in kidney disease management, where affordable POC diagnostics could transform patient outcomes in resource-limited settings worldwide.

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Appendix A

BILL OF MATERIALS AND PCB SPECIFICATIONS

A.1 Bill of Materials

Table A.1 lists all components used in the potentiostat PCB assembly, including designators, part numbers, suppliers, and unit prices as of October 2025 (sourced from LCSC Electronics via JLCPCB).

Table A.1: Complete Bill of Materials for the Potentiostat PCB.

ID	Name / Value	Desig.	LCSC Part	Qty	Price (\$)
1	MIC5205-3.3YM5 (LDO 3.3 V)	3.3V_LDO_ _2	C37970	2	0.365
2	9 V Battery connector	BT1	C703374	1	0.103
3	4.7 μ F cap (0805)	C1, C2	C7393930	2	0.016
4	10 μ F cap (0805)	C3,C6,C9,C10,C12	C82705	5	0.115
5	470 pF cap (0805)	C4, C5	C417152	2	0.060
6	100 nF cap (0805)	C7,C8,C11,C15,C16	C420082	6	0.008
7	10 nF cap (0805)	C12, C13	C7503490	2	0.005
8	AD5667RBRMZ-2REEL7 (16-bit DAC)	DAC	C650006	1	15.353
9	MAX4643EUA+ (Dual SPDT switch)	DUAL_SWITCH	C146042	1	4.384
10	HDR-M-2.54_1x1 (test point)	J1–J6	C81276	6	0.007
11	LED 0603 Red	LED1, LED2	C72044	2	0.021
12	470 Ω resistor (0805)	R1	C385392	1	0.020
13	2 k Ω resistor (0603)	R2	C209226	1	0.003

(Table A.1 continued)

ID	Name / Value	Desig.	LCSC Part	Qty	Price (\$)
14	10 k Ω resistor (0603)	R3,R4,R5,R6,R7	C2089345	5	0.010
15	1.2 k Ω resistor (0603)	R7	C2984430	1	0.023
16	16.2 k Ω resistor (0805)	R8	C228332	1	0.005
17	845 Ω resistor (0805)	R9	C218710	1	0.008
18	29.4 k Ω resistor (0805)	R10	C228690	1	0.004
19	8.06 k Ω resistor (0805)	R11	C19006	1	0.002
20	220 Ω resistor (0603)	R13	C138284	1	0.002
21	TPMAX4644EUT+TSINGLE_SWITCH (Single SPDT switch)	U1	C1241903	1	0.335
22	DS1020-03ST1D (SPE connector)	SPE_CONN	C40878053	1	—
23	SK-12D02-VG5 (Slide switch)	SW1	C136722	1	0.074
24	POWER-JACK (DC barrel jack)	U1	C990000179	1	—
25	MCP73831T-2ATI/OT (LiPo charger)	U3	C14879	1	1.701
26	ESP32-WROOM-32E 4 MB	U6	C701341	1	3.850
27	AD8608ARUZ (Quad precision op-amp)	U11	C578776	1	4.170

A.2 PCB Specifications

Table A.2: PCB fabrication specifications (JLCPCB, October 2025).

Parameter	Specification
Number of layers	2
Material	FR4
Board thickness	1.6 mm
Copper thickness	1 oz (35 μm)
Surface finish	HASL (Lead-free)
Solder mask colour	Green
Silkscreen colour	White
Minimum trace width	0.1 mm
Minimum trace spacing	0.1 mm
Minimum via drill diameter	0.3 mm
Board dimensions	Approx. 80 $\times$ 60 mm
Quantity ordered	5
Unit price (5 pcs)	\$2 (approx. Rs170 per board)